SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

COURSE CODE: ECA04

COURSE NAME: ANALOG CIRCUITS

MASTER RECORD

LIST OF EXPERIMENTS

LIST OF EXPERIMENTS	DURATION (EXPT CONDUCTION + EVALUATION)	CO	BL	WK	TYPE OF EXPT
1.Design of Voltage Series Feedback Amplifiers using BJT 1.1 Verification of Closed Loop Voltage Gain 1.2 Verification of Open Loop Voltage Gain 1.3 Frequency Response Analysis	120 Minutes	CO1	BL2 BL3 BL4	WK2 WK3 WK4 WK6	HARDWARE
2.Design of Voltage Shunt Feedback Amplifiers using BJT 2.1 Verification of Closed Loop Voltage Gain 2.2 Verification of Open Loop Voltage Gain 2.3 Frequency Response Analysis	120 Minutes	CO1	BL2 BL3 BL4	WK2 WK3 WK4 WK6	HARDWARE
3.Design and Analysis of RC Phase Shift Oscillator Circuit using BJT 3.1 Analysis of RC Phase Shift Oscillator 3.2 Percentage frequency error analysis of RC phase Oscillator	60 Minutes	CO2	BL3 BL4 BL5	WK2 WK3 WK4 WK6	HARDWARE
4.Design and Analysis of LC (Colpitts) Oscillator Circuit using BJT 4.1 Analysis of LC Colpitts Oscillator 4.2 Percentage frequency error analysis of LC Oscillator	60 Minutes	CO2	BL3 BL4 BL5	WK2 WK3 WK4 WK6	HARDWARE
5. Design and Simulation of UJT Relaxation Oscillator using LT Spice simulator 5.1 Analysis of UJT Relaxation Oscillator. 5.2 Determine Time Period and frequency 5.3 Determine Duty Cycle	60 Minutes	CO2	BL3 BL4 BL5	WK2 WK3 WK4 WK6	SIMULATION

6.Frequency Response of CB Amplifier using LT Spice Simulator 6.1 Analysis of Voltage Gain 6.2 Input and Output Characteristic Analysis 6.3 Frequency Response Analysis	60 Minutes	CO3	BL2 BL3 BL4	WK2 WK3 WK4 WK6	SIMULATIO N
7. Design and Simulation of FET Amplifier using LT Spice Simulator 7.1 Analysis of Voltage Gain 7.2 Input and Output Characteristic Analysis 7.3 Frequency Response Analysis	60 Minutes	CO3	BL2 BL3 BL4	WK2 WK3 WK4 WK6	SIMULATIO N
8. Design of Astable Multivibrator using BJT 8.1Switching Analysis of Astable Multivibrator 8.2Measurement of Time Period and Frequency 8.3 Duty Cycle Analysis	120 Minutes	CO4	BL2 BL3 BL4	WK2 WK3 WK4 WK6	HARDWARE
9. Design and Simulation of Schmitt Trigger using LT Spice Simulator 9.1 Analyze function and determine frequency and duty Cycle, 9.2 Determination Of UTP and LTP Frequency Response 9.3 Analysis of Hysteresis Characteristics	60 Minutes	CO4	BL2 BL3 BL4	WK2 WK3 WK4 WK6	SIMULATIO N
10. Design of Hartley Oscillator using PCB Software 10.1 Design and Create PCB layout for Hartley Oscillator circuit. 10.2 Route PCB tracks with Manufacturing board 10.3 Generate fabrication files.	60 Minutes	CO5	BL4 BL5 BL6	WK5 WK6 WK8	SIMULATIO N
11.Design of Monostable Multivibrator using PCB Software 11.1Design and Create PCB layout Monostable for Multivibrator circuit.	60 Minutes	CO5	BL4 BL5 BL6	WK5 WK6 WK8	SIMULATIO N

11.2 Route PCB tracks with Manufacturing board, 11.3 Generate fabrication files.					

EXPERIMENT – 1

DESIGN AND ANALYSIS OF VOLTAGE SERIES FEEDBACK AMPLIFIER USING BJT

OBJECTIVES:

The experiment is divided into three engineering laboratory model tests:

1 Verification of Closed Loop Voltage Gain with feedback
2 Verification of Closed Loop Voltage Gain without feedback
3 Frequency Response Analysis for with feedback and without feedback amplifier

EXPERIMENT – 1.1

VERIFICATION OF CLOSED-LOOP VOLTAGE GAIN (WITH FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine the closed-loop voltage gain experimentally.
2. Calculate feedback factor and loop gain.
3. Calculate Midband Frequency
4. Analyze the obtained values based on theoretical calculations and practical observations.

AIM:

To design and verify the closed-loop voltage gain of a voltage series feedback amplifier using BJT.

THEORETICAL BACKGROUND

A voltage series feedback amplifier is a negative feedback amplifier in which a fraction of the output voltage is fed back in series with the input signal. The feedback voltage opposes the input signal and stabilizes the amplifier gain.

DESIGN FORMULA:

For a voltage series feedback amplifier,

Closed-Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

Feedback Factor

$$\beta = \frac{A - A_f}{A \times A_f}$$

Loop Gain

$$A\beta = \frac{A}{A_f} - 1$$

Percentage Reduction in Gain

$$\%\ \text{Reduction in Gain} = \frac{A - A_f}{A} \times 100$$

Input Resistance with Feedback

$$R_{in,f} = R_{in}(1 + A\beta)$$

Output Resistance with Feedback

$$R_{out,f} = \frac{R_{out}}{1 + A\beta}$$

Bandwidth

$$BW = f_H - f_L$$

Gain-Bandwidth Product

$$A \times BW = A_f \times BW_f$$

Open-Loop Voltage Gain

$$A = \frac{V_o}{V_i}$$

Closed-Loop Voltage Gain (Experimental)

$$A_f = \frac{V_{of}}{V_i}$$

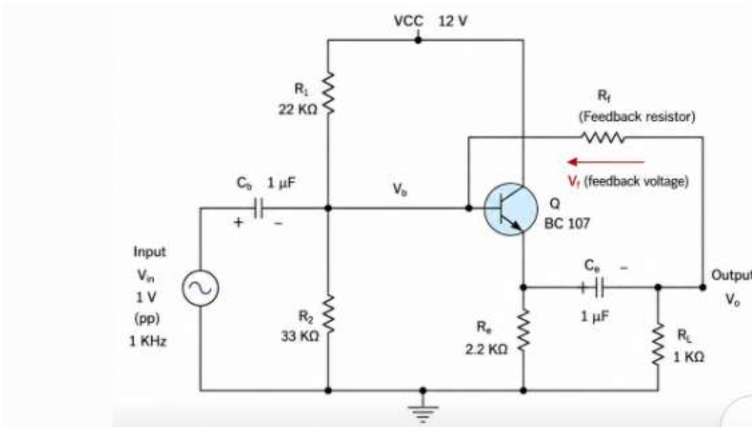
where:

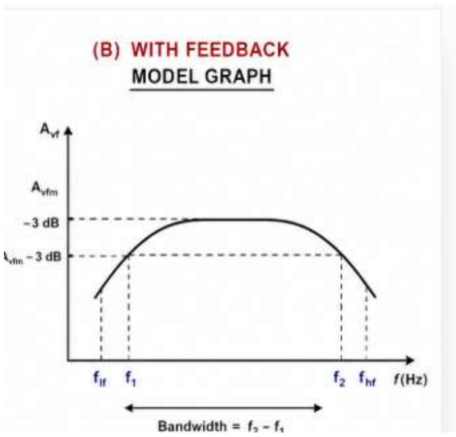
- A = Open-loop gain
- A_f = Closed-loop gain
- β = Feedback factor
- R_{in} = Input resistance without feedback
- $R_{in,f}$ = Input resistance with feedback
- R_{out} = Output resistance without feedback
- $R_{out,f}$ = Output resistance with feedback
- f_L = Lower cutoff frequency
- f_H = Upper cutoff frequency
- BW = Bandwidth
- V_i = Input voltage
- V_o = Output voltage without feedback
- V_{of} = Output voltage with feedback

EXPERIMENTAL PROCEDURE:

1. Verify all components using a mustimeter.
2. Assemble the voltage series feedback amplifier circuit on a breadboard.
3. Apply a DC supply voltage of 10 V.
4. Initially disconnect the feedback resistor.
5. Apply a sinusoidal input signal of 20 mV.
6. Measure input voltage and output voltage.
7. Calculate open-loop gain.
8. Connect the feedback network.
9. Measure input voltage and output voltage again.
10. Calculate closed-loop gain.
11. Determine feedback factor.
12. Compare theoretical and practical values.
13. Record all observations.

CIRCUIT DIAGRAM





TABULATION 1

Measurement of frequency response of voltage series with feedback amplifiers .

Vin=

Frequency (in Hz)	V_0 (Volts)	Gain= V_0/V_{in}	Gain (dB) = 20 log(V_o/V_{in})

RESULT

The closed-loop voltage gain of the voltage series with feedback amplifier was verified successfully.

EXPERIMENT – 1.2

VERIFICATION OF OPEN-LOOP VOLTAGE GAIN (WITHOUT FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine the open-loop voltage gain experimentally.
2. Calculate Midband Frequency and voltage gain
3. Analyze the obtained values based on theoretical calculations and practical observations.

AIM:

To design and verify the open-loop voltage gain of a voltage series without feedback amplifier using BJT.

THEORETICAL BACKGROUND

A **BJT amplifier without feedback**, also known as an **open-loop amplifier**, amplifies the input signal without feeding any portion of the output back to the input. The voltage gain of the amplifier depends entirely on the transistor characteristics and the external circuit components such as resistors and the supply voltage. In an open-loop amplifier, the input signal is applied directly to the transistor, and the amplified output is obtained from the collector. Since there is no feedback network, the amplifier provides a high voltage gain. However, the gain is sensitive to changes in transistor parameters, temperature, and power supply variations, resulting in poor gain stability.

Expected Learning Outcomes

After completing this experiment, students will be able to:

1. Design and construct a BJT amplifier without feedback (open-loop amplifier).
2. Measure the input and output voltages of the amplifier.
3. Determine the open-loop voltage gain experimentally.
4. Analyze the frequency response of the amplifier.
5. Compare the theoretical and practical values of the voltage gain.

6. Understand the limitations of an amplifier without feedback, such as poor gain stability, higher distortion, and limited bandwidth.
7. Interpret the experimental results and relate them to the characteristics of an open-loop amplifier.

Formula:

For a **BJT amplifier without feedback (Open-Loop Amplifier)**, the commonly used formulas are:

1. Voltage Gain (Open-Loop)

$$\boxed{A_v = \frac{V_o}{V_{in}}}$$

Where:

- A_v = Voltage gain
- V_o = Output voltage (V)
- V_{in} = Input voltage (V)

2. Voltage Gain in Decibels (dB)

$$\boxed{Gain\,(dB) = 20\log_{10}\left(\frac{V_o}{V_{in}}\right)}$$

3. Input Resistance

$$\boxed{R_{in} = \frac{V_{in}}{I_{in}}}$$

Where:

- R_{in} = Input resistance (Ω)
- I_{in} = Input current (A)

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4. Output Resistance

$$\boxed{R_{out} = \frac{V_{out}}{I_{out}}}$$

Where:

- R_{out} = Output resistance (Ω)
- I_{out} = Output current (A)

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5. Bandwidth

$$\boxed{BW = f_H - f_L}$$

Where:

- f_H = Upper cutoff frequency (Hz)
- f_L = Lower cutoff frequency (Hz)

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6. Current Gain (if required)

$$\boxed{A_i = \frac{I_o}{I_{in}}}$$

7. Power Gain (if required)

$$\boxed{A_p = A_v \times A_i}$$

or

$$\boxed{A_p = \frac{P_o}{P_{in}}}$$

EXPERIMENTAL PROCEDURE

1. Verify all the components and the BJT transistor using a multimeter.
2. Assemble the BJT amplifier circuit on a breadboard as per the circuit diagram, ensuring that the feedback resistor is disconnected (without feedback configuration).
3. Connect the DC power supply and set the supply voltage to **10 V**.
4. Connect the function generator to the amplifier input and apply a sinusoidal input signal of **20 mV** at **1 kHz**.
5. Measure the input voltage ((V_{in})) using the CRO.
6. Measure the output voltage ((V_{o})) using the CRO.
7. Record the output voltage at each frequency.

TABULATION:

Measurement of frequency response of voltage series without feedback amplifiers .

Vin=

Frequency (in Hz)	**V_0 (Volts)**	**Gain= V_0/V_{in}**	**Gain (dB) = 20 log(V_o/V_{in})**

RESULT

The open-loop voltage gain of the voltage series without feedback amplifier was verified successfully.

EXPERIMENT – 1.3

FREQUENCY RESPONSE ANALYSIS

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine lower and upper cutoff frequencies.
2. Evaluate bandwidth improvement due to feedback.

Aim

To obtain the frequency response of a voltage series feedback amplifier and determine its bandwidth.

THEORY

Negative feedback improves amplifier bandwidth while reducing gain.

The gain-bandwidth product remains approximately constant.

Gain:

$$A_v = \frac{V_o}{V_i}$$

Gain in dB:

$$Gain(dB) = 20\log_{10}\left(\frac{V_o}{V_i}\right)$$

Bandwidth:

$$BW = f_H - f_L$$

Where:

fH = Upper cutoff frequency

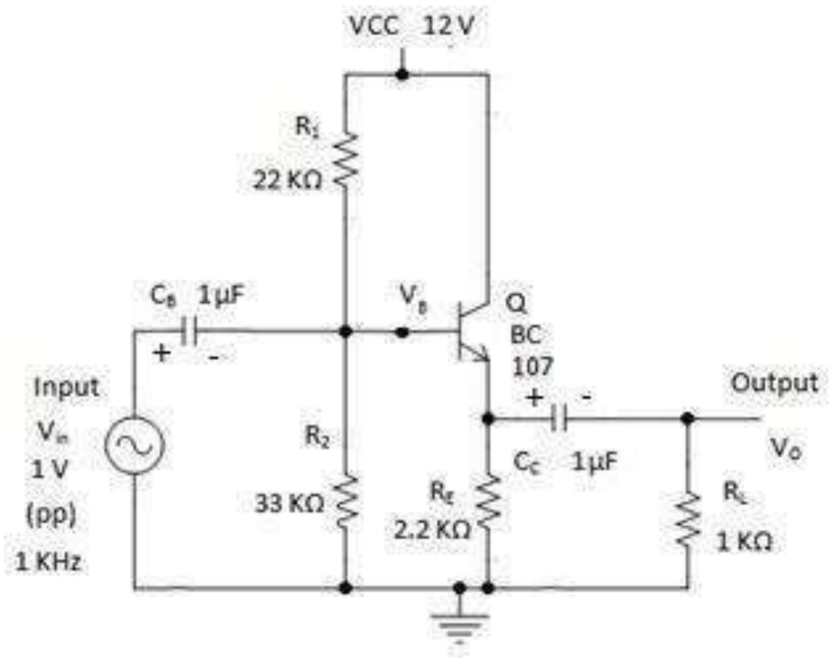
fL = Lower cutoff frequency

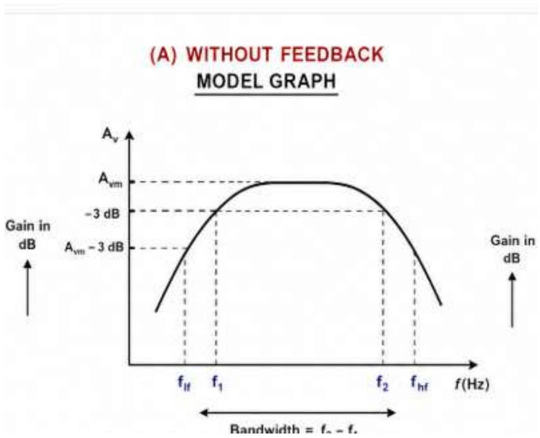
DETAILED PROCEDURE

1. Assemble the voltage series feedback amplifier.
2. Apply a constant input signal of 20 mV.
3. Set frequency to 100 Hz.
4. Measure output voltage.
5. Increase frequency gradually.

6. Record output voltage for each frequency.
7. Calculate gain.
8. Convert gain into dB.
9. Plot frequency response curve.
10. Determine cutoff frequencies.
11. Calculate bandwidth.

CIRCUIT DIAGRAM





VERIFICATION TABLE

Parameter	CLOSED LOOP VOLTAGE	OPEN LOOP VOLTAGE
Lower Cutoff Frequency		
Upper Cutoff Frequency		
Band width		

RESULT

The frequency response characteristics of the voltage series feedback amplifier with feedback and without feedback (closed loop and open loop) were plotted successfully.

EXPERIMENT – 2

DESIGN AND ANALYSIS OF VOLTAGE SHUNT FEEDBACK AMPLIFIER USING BJT

OBJECTIVES:

The experiment is divided into three engineering laboratory model tests:

1.Verification of Closed Loop Voltage Gain

2. Verification of Open Loop Voltage Gain

3.Frequency Response Analysis

EXPERIMENT – 2.1

CLOSED-LOOP VOLTAGE GAIN (WITH FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design a voltage shunt feedback amplifier using BJT.
2. Measure open-loop and closed-loop gains.
3. Determine feedback factor and loop gain.
4. Analyze the effect of shunt mixing on amplifier performance.
5. Compare theoretical and practical results.

AIM:

To design and verify the closed-loop voltage gain of a voltage shunt feedback amplifier using BJT.

THEORETICAL BACKGROUND

A voltage shunt feedback amplifier samples the output voltage and feeds a fraction of it back in parallel (shunt) with the input signal.

Because the feedback signal is connected in parallel with the input:

- Gain decreases
- Input resistance decreases
- Output resistance decreases
- Bandwidth increases
- Linearity improves

The closed-loop gain is given by

DESIGN FORMULA:

For a voltage shuntfeedback amplifier,

$$A_f = \frac{A}{1 + A\beta}$$

Where:

A = Open-loop gain

A_f = Closed-loop gain

β = Feedback factor

$A\beta$ = Loop gain

Closed Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

Feedback Factor

$$\beta = \frac{V_f}{V_o}$$

Loop Gain

$$A\beta = A \times \beta$$

Midband Gain (Closed-Loop)

$$A_{f(mid)} = \frac{V_{of}}{V_i}$$

(i) Closed-Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

(ii) Input Resistance with Feedback

For Voltage Shunt Feedback:

$$R_{in,f} = \frac{R_{in}}{1 + A\beta}$$

(iii) Output Resistance with Feedback

For Voltage Shunt Feedback:

$$R_{out,f} = \frac{R_{out}}{1 + A\beta}$$

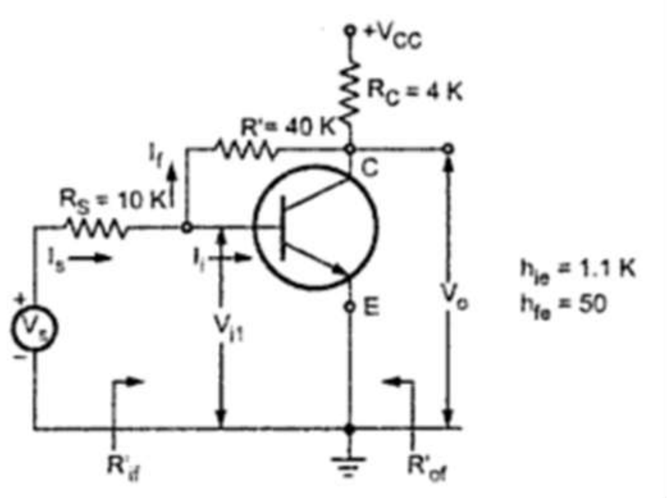
(iv) Percentage Reduction in Gain

$$\% \text{ Reduction in Gain} = \frac{A - A_f}{A} \times 100$$

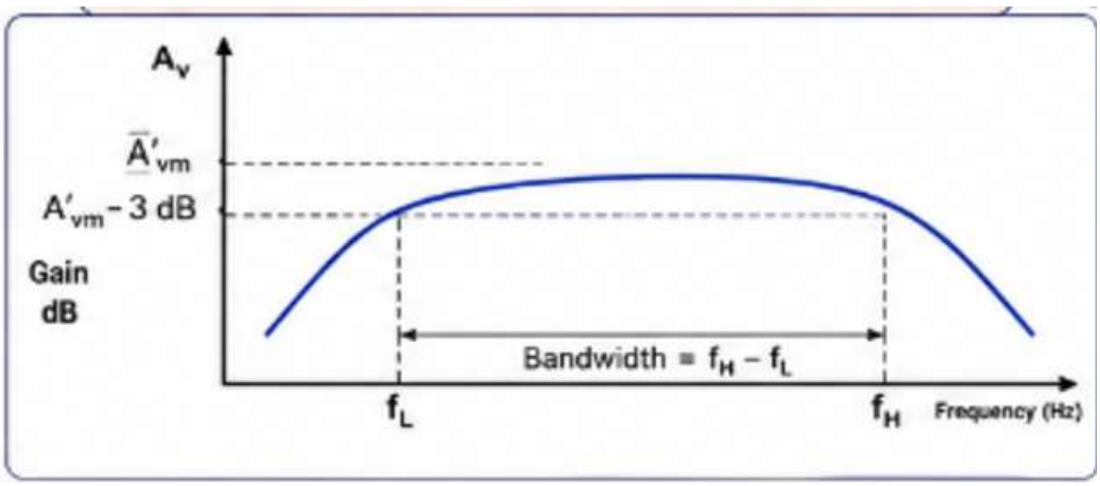
EXPERIMENTAL PROCEDURE:

1. EXPERIMENTAL PROCEDURE
2. Verify all components using a multimeter.
3. Assemble the voltage shunt feedback amplifier circuit.
4. Apply a DC supply voltage of 10 V.
5. Initially disconnect the feedback resistor.
6. Apply a sinusoidal input signal of 20 mV.
7. Measure input and output voltages.
8. Calculate open-loop gain.
9. Connect the feedback network.
10. Measure output voltage again.
11. Calculate closed-loop gain.
12. Determine feedback factor.
13. Compare theoretical and practical values.
14. Record observations.

CIRCUIT DIAGRAM



Model Graph:



TABULATION

Measurement of frequency response of voltage shunt feedback amplifiers .

Vin=

Frequency (in Hz)	V_0 (Volts)	Gain= V_0/V_{in}	Gain (dB) = 20 log(V_o/V_{in})

RESULT

The closed-loop voltage gain of the voltage shunt feedback amplifier was verified successfully.

EXPERIMENT – 2.2

VERIFICATION OF OPEN LOOP VOLTAGE GAIN

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Understand the basic working principle of a **shunt amplifier** and its open-loop configuration
2. Identify the difference between **open-loop and closed-loop amplifier operation**
3. Measure input and output voltages using CRO/DSO and DMM accurately
4. Calculate **open-loop voltage gain** using experimental readings
5. Analyze how amplification behaves without feedback and recognize its limitations
6. Develop skills in setting up and troubleshooting analog amplifier circuits
7. Understand the importance of feedback in improving amplifier stability and performance
8. Compare theoretical and practical gain values and interpret variations

Aim

To verify the open-loop voltage gain of a shunt amplifier by measuring the input and output voltages.

THEORETICAL BACKGROUND

A **shunt amplifier** is an amplifier configuration in which the feedback is connected in parallel (shunt) with the input. Before applying feedback, the amplifier operates in **open-loop mode**, where the output depends only on the intrinsic gain of the amplifier. The **open-loop voltage gain** is the ratio of the output voltage to the input voltage without any feedback network connected. Measuring the open-loop gain helps understand the amplifier's performance before studying the effects of feedback. Open-loop gain is usually high, but it is less stable and more sensitive to component variations and noise. Negative feedback is later used to improve stability, bandwidth, and linearity.

Open-Loop Voltage Gain

$$A_v = \frac{V_{out}}{V_{in}}$$

where:

- A_v = Open-loop voltage gain
- V_{out} = Output voltage (V)
- V_{in} = Input voltage (V)

Gain in Decibels (Optional)

$$A_v(\mathrm{dB}) = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

EXPERIMENTAL PROCEDURE

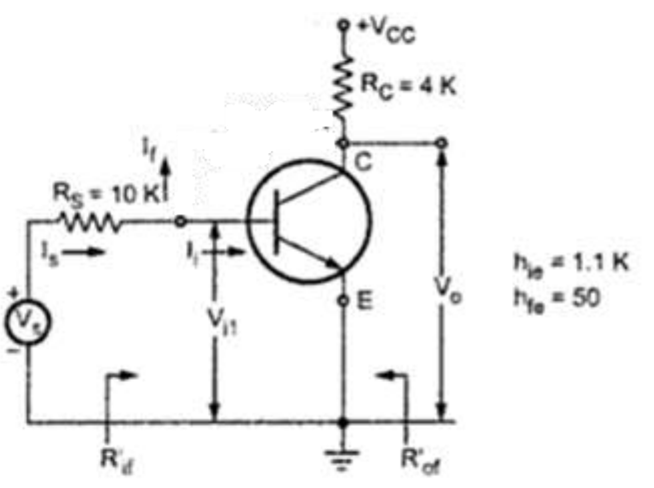
1. Connect the circuit according to the open-loop shunt amplifier circuit diagram.
2. Ensure the feedback connection is removed.
3. Switch ON the regulated DC power supply.
4. Apply a sinusoidal input signal (typically **100 mV to 200 mV** at **1 kHz**) from the function generator.
5. Measure the input voltage (VinV_{in}Vin) using the CRO/DSO.
6. Measure the output voltage (VoutV_{out}Vout).
7. Record the readings in the observation table.
8. Calculate the open-loop voltage gain using the formula.
9. Compare the calculated value with the theoretical value.

TABULATION

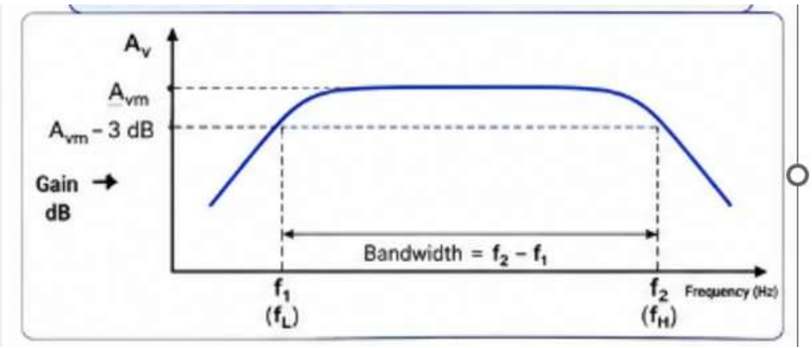
Measurement of frequency response of voltage shunt feedback amplifiers .

Circuit Diagram:

Model graph:



Model Graph:



Vin=

Frequency (in Hz)	V_0 (Volts)	Gain= V_0/V_{in}	Gain (dB) = 20 log(V_o/V_{in})

RESULT

The open-loop voltage gain of the shunt amplifier was determined by measuring the input and output voltages and verified using:

EXPERIMENT – 2.3

FREQUENCY RESPONSE ANALYSIS

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot frequency response characteristics.
2. Calculate gain in decibels.
3. Determine amplifier bandwidth.
4. Compare bandwidth with and without feedback.

Aim

To obtain the frequency response of a voltage shunt feedback amplifier and determine its bandwidth.

THEORY

Negative feedback improves amplifier bandwidth while reducing gain.

The gain-bandwidth product remains approximately constant.

Gain:

$$A_v = \frac{V_o}{V_i}$$

Gain in dB:

$$Gain(dB) = 20\log_{10}\left(\frac{V_o}{V_i}\right)$$

Bandwidth:

$$BW = f_H - f_L$$

Where:

fH = Upper cutoff frequency

fL = Lower cutoff frequency

(i) Feedback Factor

From

$$A_f = \frac{A}{1 + A\beta}$$

Therefore,

$$\beta = \frac{A - A_f}{A \times A_f}$$

(ii) Loop Gain

$$A\beta = \frac{A}{A_f} - 1$$

(iii) New Input Resistance

$$R_{in,f} = \frac{R_{in}}{1 + A\beta}$$

(iv) Feedback Resistor Network

For a resistive voltage divider:

$$\beta = \frac{R_1}{R_1 + R_f}$$

or

$$R_f = R_1 \left(\frac{1 - \beta}{\beta}\right)$$

where:

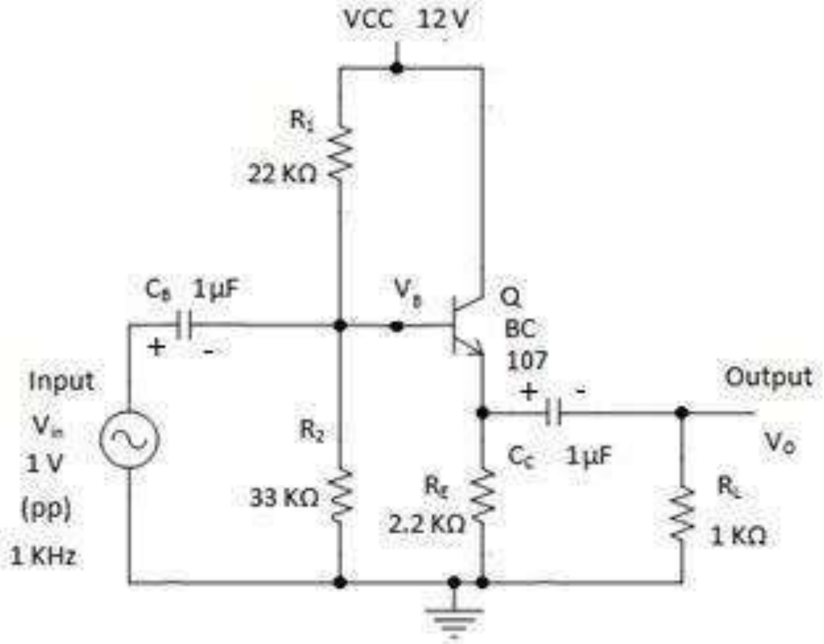
- R_f = Feedback resistor
- R_1 = Input-side resistor of feedback network

A	Open-Loop Voltage Gain (Gain without feedback)
A_f	Closed-Loop Voltage Gain (Gain with feedback)
β	Feedback Factor
$A\beta$	Loop Gain (Return Difference)
R_{in}	Input Resistance without Feedback
$R_{in,f}$	Input Resistance with Feedback
R_{out}	Output Resistance without Feedback
$R_{out,f}$	Output Resistance with Feedback
R_{of}	Effective Output Resistance with Feedback
R_L	Load Resistance
R_f	Feedback Resistor
R_1	Input-side Resistor in Feedback Network
D	Distortion without Feedback
D_f	Distortion with Feedback
V_i	Input Voltage
V_o	Output Voltage without Feedback
V_{of}	Output Voltage with Feedback
I_i	Input Current
I_o	Output Current

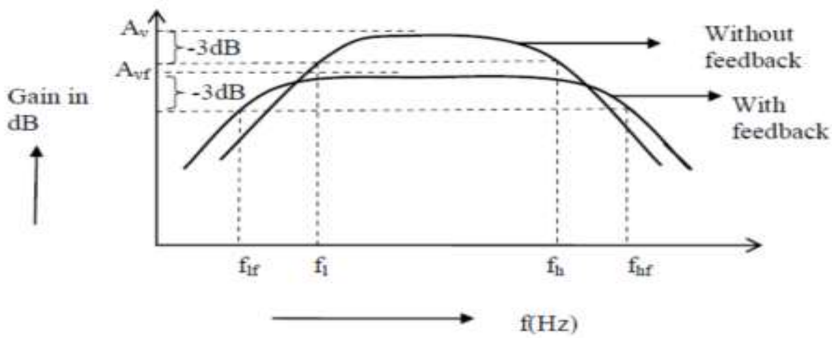
DETAILED PROCEDURE

1. Connect the circuit without feedback.
2. Apply a constant input signal of 20 mV.
3. Measure output voltage at various frequencies.
4. Calculate gain and gain in dB.
5. Connect feedback resistor RF.
6. Repeat measurements.
7. Plot frequency response curves.
8. Determine cutoff frequencies.
9. Calculate bandwidth.

CIRCUIT DIAGRAM



MODEL GRAPH :



Bandwidth= f_l-f_h

TABULATION

Parameter	CLOSED LOOP VOLTAGE	OPEN LOOP VOLTAGE
Lower Cutoff Frequency		
Upper Cutoff Frequency		
Band width		

RESULT

The frequency response characteristics of the voltage shunt feedback amplifier were plotted successfully and the bandwidth improvement due to feedback was verified.

EXPERIMENT – 3

DESIGN AND ANALYSIS OF RC PHASE SHIFT OSCILLATOR USING BJT

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1.Analysis of RC Phase Shift Oscillator

2. Percentage frequency error analysis of RC phase shift Oscillator

EXPERIMENT – 3.1

ANALYSIS OF RC PHASE SHIFT OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an RC Phase Shift Oscillator using BJT.
2. Understand Barkhausen Criterion.
3. Measure oscillation frequency.
4. Analyze feedback networks in oscillators.
5. Compare theoretical and practical oscillation frequencies.

Aim

To design and verify the operation of an RC Phase Shift Oscillator using BJT.

THEORETICAL BACKGROUND

An oscillator is an electronic circuit that generates periodic waveforms without requiring an external input signal.

An RC Phase Shift Oscillator consists of:

1. A transistor amplifier stage.
2. Three RC sections providing phase shift.
3. Positive feedback network.

The transistor amplifier provides:180°phase shift.
The RC network provides:180°additional phase shift.
Hence total phase shift becomes:360° which satisfies Barkhausen Criterion.

DESIGN FORMULA

Frequency of Oscillation

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

where

R = Resistance of each RC section

C = Capacitance of each RC section

2. Resistance Calculation

$$R = \frac{1}{2\pi f_o C\sqrt{6}}$$

3. Capacitance Calculation

$$C = \frac{1}{2\pi f_o R\sqrt{6}}$$

4. Condition for Sustained Oscillations (Barkhausen Criterion)

$$A\beta = 1$$

5. Feedback Factor of Three-Section RC Network

$$\beta = \frac{1}{29}$$

6. Minimum Amplifier Gain Required

$$A_{min} = 29$$

7. Gain in Decibels

$$G_{dB} = 20\log_{10}(A)$$

8. Gain Margin

$$\text{Gain Margin} = 20\log_{10}\left(\frac{A}{A_{min}}\right)$$

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9. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

10. Base Current

$$I_B = \frac{I_C}{\beta}$$

---

11. Collector Current

$$I_C = \beta I_B$$

---

12. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

13. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

14. Base Voltage

$$V_B = V_E + V_{BE}$$

15. Voltage Divider Bias Resistors

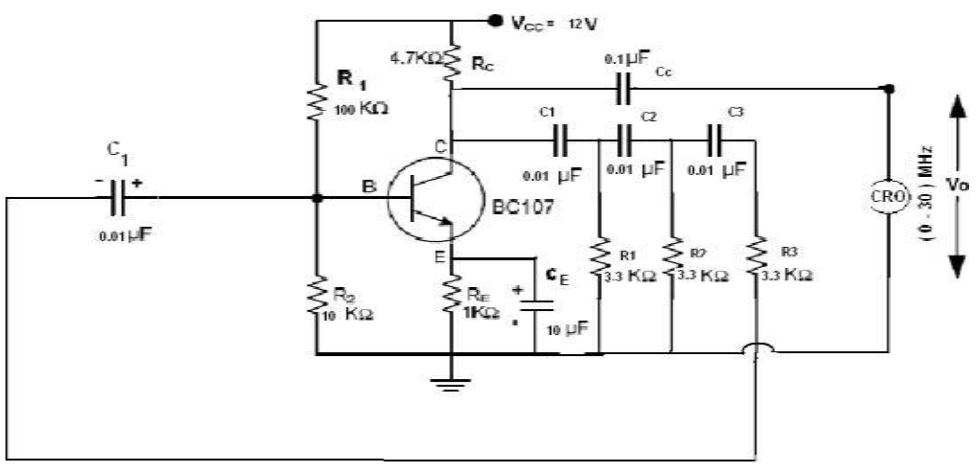
$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

$$R_2 = \frac{V_B}{I_{R2}}$$

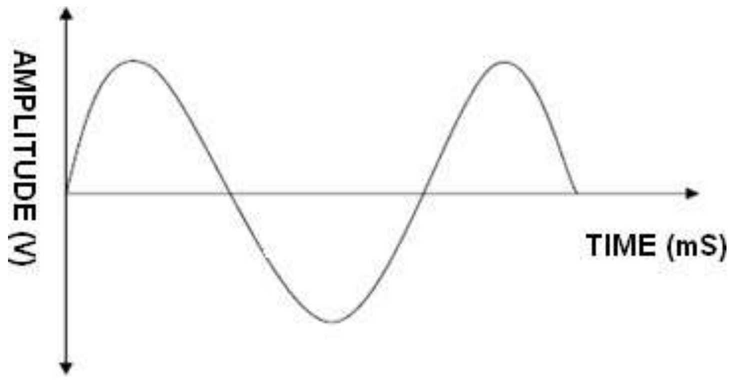
Procedure:

1. Verify all components.
2. Assemble RC phase shift oscillator circuit.
3. Apply DC supply voltage.
4. Verify transistor biasing.
5. Observe output waveform on CRO.
6. Measure oscillation frequency.
7. Compare theoretical and practical values.
8. Record observations.

CIRCUIT DIAGRAM:



Model Graph:



Tabulation:

Oscillator	RC PHASE SHIFT OSCILLATOR
AMPLITUDE (V)	
TIME PERIOD (ms)	
FREQUENCY (Hz)	

RESULT

The RC Phase Shift Oscillator was designed successfully and the oscillation frequency was verified.

EXPERIMENT – 3.2

PERCENTAGE FREQUENCY ERROR ANALYSIS OF RC PHASE SHIFT OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Compare RC Oscillators.
2. Analyze frequency stability.
3. Evaluate oscillator performance in terms of Percentage Frequency Error.

Aim

To analyze the percentage frequency error of RC phase shift and LC oscillator

THEORY

- RC Oscillator
- Suitable for audio frequencies.
- Uses resistors and capacitors.
- Simple construction.
- Better frequency stability.

DESIGN FORMULA:

RC Oscillator

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

EXPERIMENTAL PROCEDURE

Procedure

1. Connect the RC phase shift oscillator circuit correctly.
2. Verify all resistor and capacitor values.
3. Apply the required DC supply voltage.
4. Observe the output waveform on the CRO.
5. Adjust the circuit (if necessary) until a stable sine wave is obtained.
6. Measure the output frequency using the frequency counter or CRO.
7. Calculate the theoretical frequency using frequency formula
8. Compare the measured and theoretical frequencies.
9. Calculate the percentage frequency error.

Formula

Theoretical Frequency

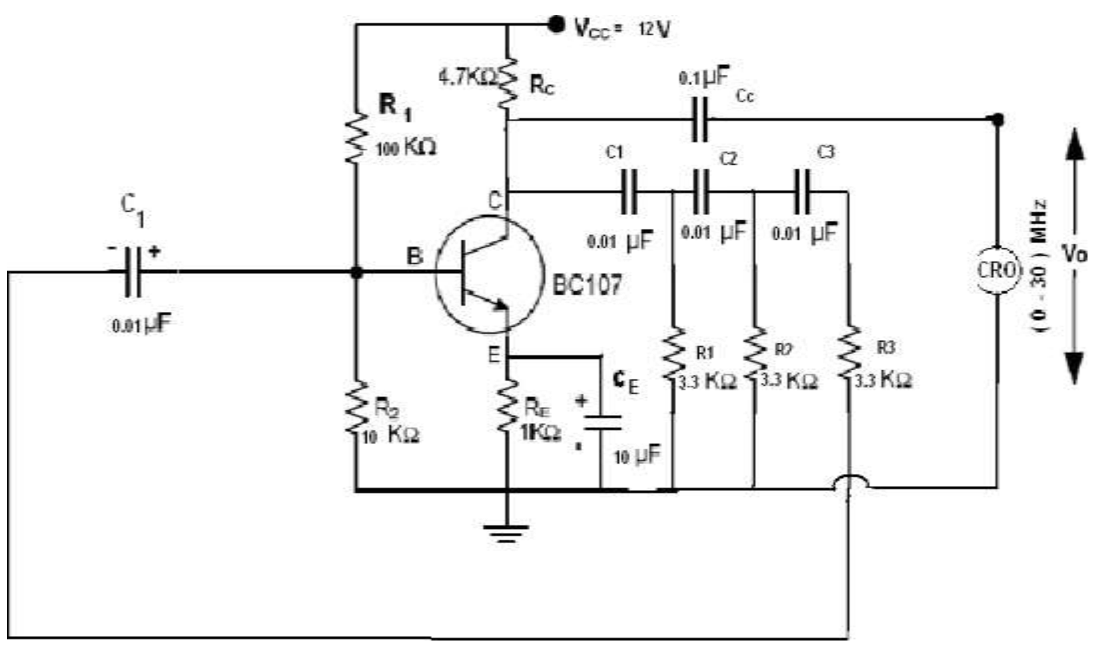
$$f_t = \frac{1}{2\pi RC\sqrt{6}}$$

Percentage Frequency Error

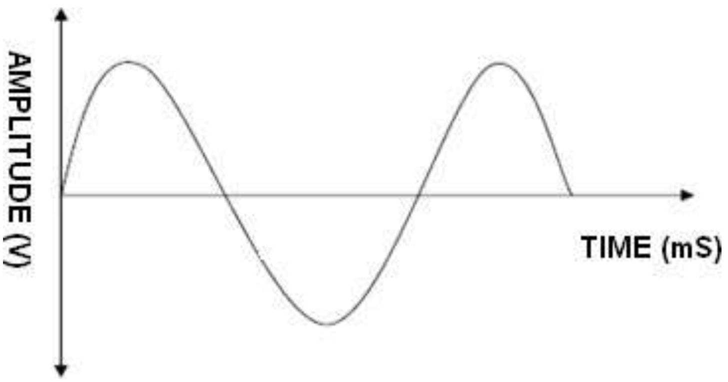
$$\%\ \text{Error} = \frac{|f_m - f_t|}{f_t} \times 100$$

Where:

- f_t = Theoretical frequency
- f_m = Measured frequency



Model Graph:



Tabulation:

RC Oscillator	THEORITICAL	PRACTICAL
Frequency (kHz)		
% Error/ Stability		

RESULT

The percentage frequency error of RC and LC oscillators were analyzed successfully. The LC oscillator exhibited superior frequency stability compared to the RC Phase Shift Oscillator.

EXPERIMENT – 4

DESIGN AND ANALYSIS OF LC (COLPITTS) OSCILLATOR CIRCUIT USING BJT

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an LC oscillator (Colpitts).
2. Measure resonant frequency.
3. Analyze energy transfer between L and C.
4. Verify oscillation conditions.
5. Compare theoretical and practical frequencies.

Aim

To design and verify the operation of an LC oscillator (Colpitts) using BJT.

THEORETICAL BACKGROUND

LC oscillators generate sinusoidal waveforms using an inductive-capacitive tank circuit. The oscillation frequency depends on: Inductance (L) and Capacitance (C)

DESIGN FORMULA:

Resonant Frequency

$$f = \frac{1}{2\pi\sqrt{LC}}$$

where

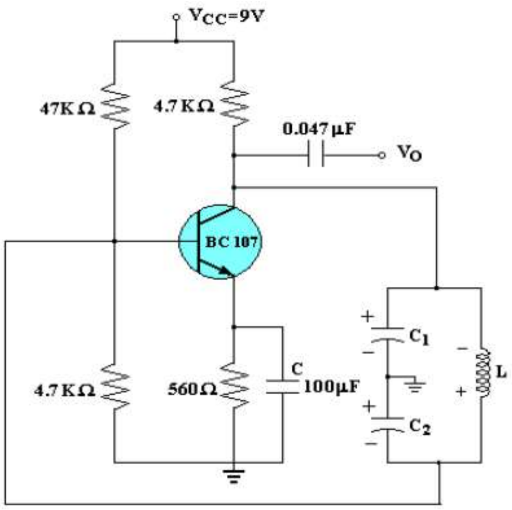
L = Inductance

C = Capacitance

EXPERIMENTAL PROCEDURE

1. Assemble LC oscillator circuit.
2. Verify component values.
3. Apply supply voltage.
4. Observe sinusoidal waveform.
5. Measure frequency.
6. Compare theoretical and practical frequencies.
7. Record observations.

Circuit diagram:

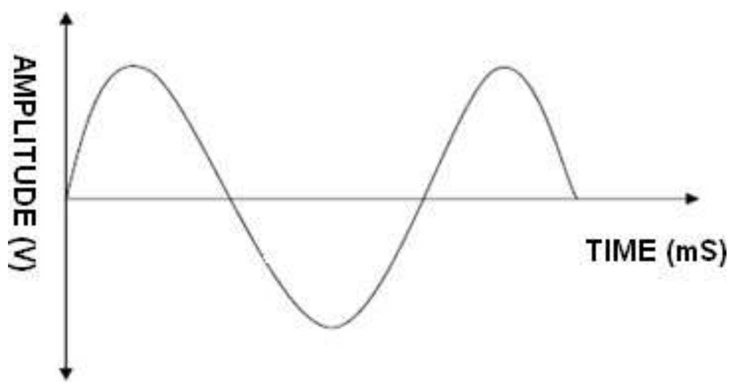


Colpitts Oscillator

Tabulation:

Oscillator	RC PHASE SHIFT OSCILLATOR
AMPLITUDE (V)	
TIME PERIOD (ms)	
FREQUENCY (Hz)	

Model Graph:



RESULT

The LC oscillator was designed successfully and the oscillation frequency was verified.

EXPERIMENT – 3.2

PERCENTAGE FREQUENCY ERROR ANALYSIS OF LC OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Compare RC and LC oscillators.
2. Analyze frequency stability.
3. Evaluate oscillator performance in terms of Percentage Frequency Error.

Aim

To analyze the percentage frequency error of LC oscillator

THEORY

- LC Oscillator
- Suitable for audio frequencies.
- Uses resistors and capacitors.
- Better frequency stability.

DESIGN FORMULA:

LC Oscillator

$$f = \frac{1}{2\pi\sqrt{LC}}$$

LC Colpitts Oscillator (BJT) – Required Formulae Only

1. Equivalent Capacitance of Colpitts Tank Circuit

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

where:

- C_1 = Capacitor 1
- C_2 = Capacitor 2
- C_T = Equivalent capacitance

2. Frequency of Oscillation

$$f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

where:

- L = Inductance
- C_T = Equivalent capacitance

---

3. Inductance Calculation

$$L = \frac{1}{(2\pi f_o)^2 C_T}$$

4. Equivalent Capacitance Calculation

$$C_T = \frac{1}{(2\pi f_o)^2 L}$$

---

5. Barkhausen Criterion

$$A\beta = 1$$

where:

- A = Amplifier Gain
- β = Feedback Factor

6. Feedback Factor

$$\beta = \frac{C_1}{C_2}$$

or

$$\beta = \frac{C_2}{C_1}$$

(depending on the feedback connection)

---

7. Minimum Gain Condition

$$A \geq \frac{C_2}{C_1}$$

---

8. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

9. Base Current

$$I_B = \frac{I_C}{\beta}$$

10. Collector Current

$$I_C = \beta I_B$$

11. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

12. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

13. Base Voltage

$$V_B = V_E + V_{BE}$$

14. Voltage Divider Bias

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

$$R_2 = \frac{V_B}{I_{R2}}$$

15. Output Voltage

$$V_{pp} = 2V_p$$

where:

- V_{pp} = Peak-to-Peak Voltage
- V_p = Peak Voltage

---

16. Frequency Stability

Percentage frequency change:

$$\%\Delta f = \frac{f_2 - f_1}{f_1} \times 100$$

17. Frequency Stability Factor

$$S_f = \frac{\Delta f}{f_o}$$

---

1. Frequency of Oscillation

$$f_o = \frac{1}{2\pi\sqrt{L_T C_T}}$$

where

$$L_T = L_1 + L_2 + 2M$$

For negligible mutual inductance,

$$L_T = L_1 + L_2$$

If two capacitors are connected in series:

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

If

$$C_1 = C_2 = C$$

then

$$C_T = \frac{C}{2}$$

3. Inductance Calculation

$$L_T = \frac{1}{(2\pi f_o)^2 C_T}$$

---

4. Capacitance Calculation

$$C_T = \frac{1}{(2\pi f_o)^2 L_T}$$

---

5. Barkhausen Criterion

$$A\beta = 1$$

6. Feedback Ratio of Hartley Oscillator

$$\beta = \frac{L_1}{L_2}$$

or

$$\beta = \frac{L_2}{L_1}$$

(depending on the feedback winding arrangement)

---

7. Minimum Gain Condition

$$A \geq \frac{L_2}{L_1}$$

or

$$A \geq \frac{L_1}{L_2}$$

---

8. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

9. Base Current

$$I_B = \frac{I_C}{\beta}$$

10. Collector Current

$$I_C = \beta I_B$$

11. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

12. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

13. Base Voltage

$$V_B = V_E + V_{BE}$$

14. Voltage Divider Bias

Upper Bias Resistor

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

Lower Bias Resistor

$$R_2 = \frac{V_B}{I_{R2}}$$

15. Output Voltage

$$V_{pp} = 2V_p$$

where

- V_{pp} = Peak-to-Peak Output Voltage
- V_p = Peak Output Voltage

EXPERIMENTAL PROCEDURE

Procedure

1. Connect the LC oscillator circuit correctly.
2. Verify the values of the inductor and capacitor.
3. Apply the specified DC supply voltage.
4. Observe the output waveform on the CRO.
5. Ensure a stable sinusoidal waveform is obtained.
6. Measure the output frequency using the frequency counter or CRO.
7. Calculate the theoretical frequency using formula
8. Compare the measured frequency with the theoretical frequency.
9. Calculate the percentage frequency error.

Formula

Theoretical Frequency

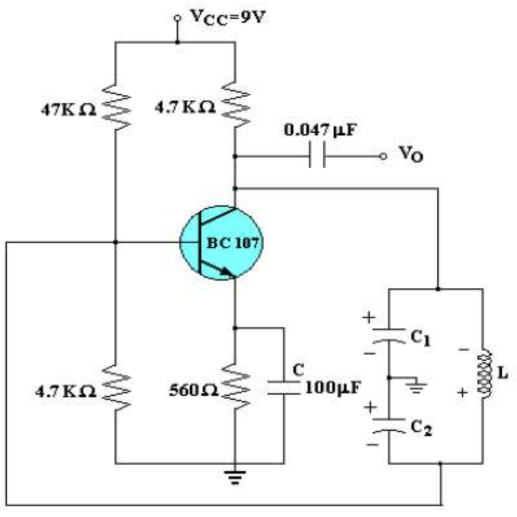
$$f_t = \frac{1}{2\pi\sqrt{LC}}$$

Percentage Frequency Error

$$\%\ \text{Error} = \frac{|f_m - f_t|}{f_t} \times 100$$

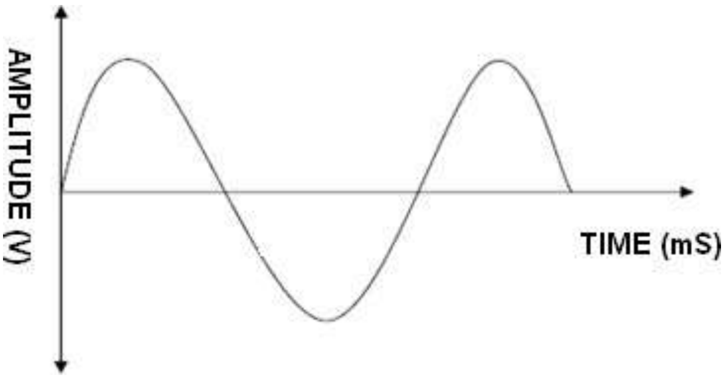
Where:

- f_t = Theoretical frequency
- f_m = Measured frequency



Colpitts Oscillator

Model Graph:



Tabulation:

RC Oscillator	THEORITICAL	PRACTICAL
Frequency (kHz)		
% Error/ Stability		

RESULT

The percentage frequency error of LC oscillators were analyzed successfully.

EXPERIMENT – 5

DESIGN AND SIMULATION OF UJT RELAXATION OSCILLATOR USING LTSPICE

OBJECTIVES

1. Analysis of UJT Relaxation Oscillator.
2. Determine Bandwidth
3. Determine Duty Cycle

EXPERIMENT – 5.1

FREQUENCY ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the construction and operating principle of a Unijunction Transistor (UJT).
2. Explain the working of a UJT Relaxation Oscillator.
3. Verify the generation of oscillations using a UJT-based oscillator circuit.
4. Observe the charging and discharging characteristics of the timing capacitor.
5. Measure the oscillation frequency and compare it with the theoretical value.
6. Analyze the effect of resistance and capacitance values on the oscillation frequency.
7. Identify and interpret the sawtooth waveform and trigger pulse waveform generated by the circuit.
8. Determine the role of the intrinsic stand-off ratio (η)(\eta)(η) in oscillator operation.
9. Plot and analyze the output waveforms using a CRO/oscilloscope or simulation tool.
10. Apply UJT oscillators in timing, triggering, pulse generation, and sweep generation applications.

Aim

To verify the operation of a UJT Relaxation Oscillator.

THEORY

The UJT relaxation oscillator generates non-sinusoidal oscillations.

Operation:

- Capacitor charges exponentially.
- UJT triggers at peak voltage.
- Capacitor discharges rapidly.
- Cycle repeats continuously.

DESIGN FORMULA:

Frequency

$$f = \frac{1}{RC \ln\left(\frac{1}{1-\eta}\right)}$$

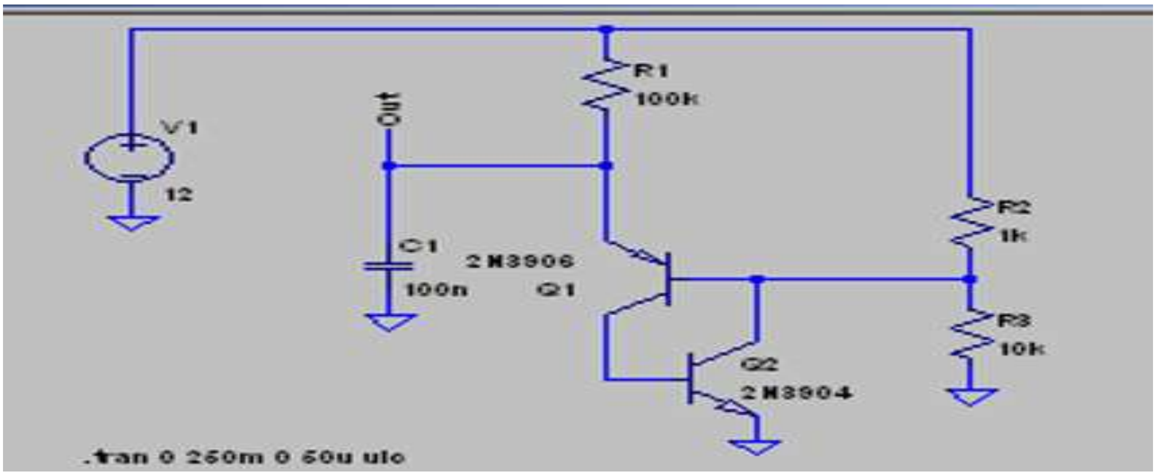
where

η = Intrinsic Stand-off Ratio

PROCEDURE

1. Draw UJT oscillator in LTSpice.
2. Configure RC network.
3. Run transient analysis.
4. Observe sawtooth waveform.
5. Measure oscillation frequency.

Circuit diagram:



<u>Procedure:</u>

1.Connections are made as shown in the circuit diagram.

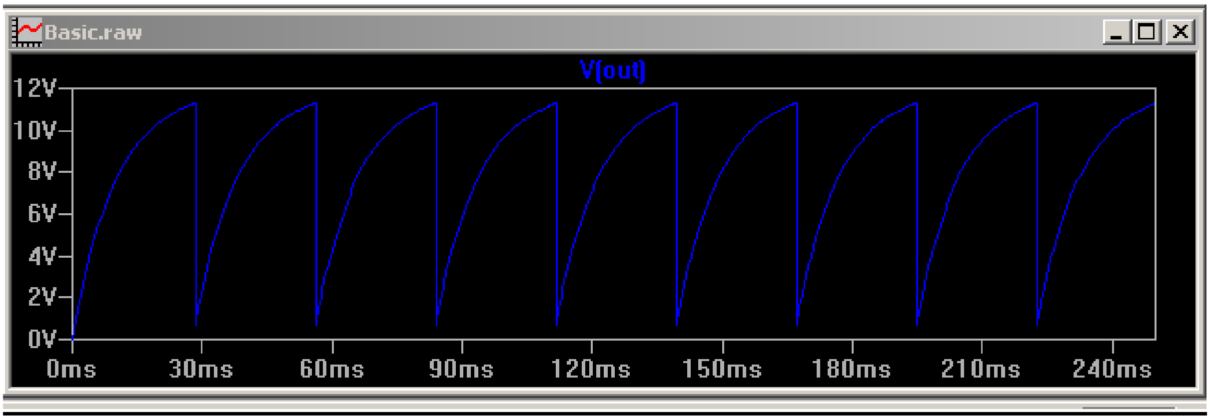
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	
Time period	
Frequency	

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 5.2

BANDWIDTH ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

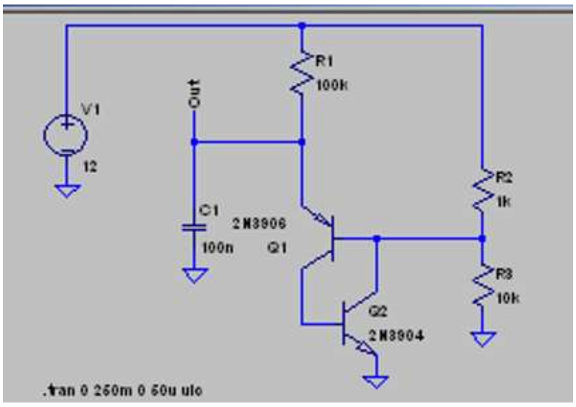
After completion of this experiment, students will be able to:

1. Understand the operating principle of a UJT Relaxation Oscillator and its frequency-determining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
6. Investigate the effect of varying resistance (R)(R)(R) and capacitance (C)(C)(C) values on the oscillation frequency.
7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
9. Apply LTSpice simulation tools for frequency analysis and performance evaluation of relaxation oscillators.

Aim

To verify frequency variation with RC values.

Circuit diagram:



Procedure:

1.Connections are made as shown in the circuit diagram.

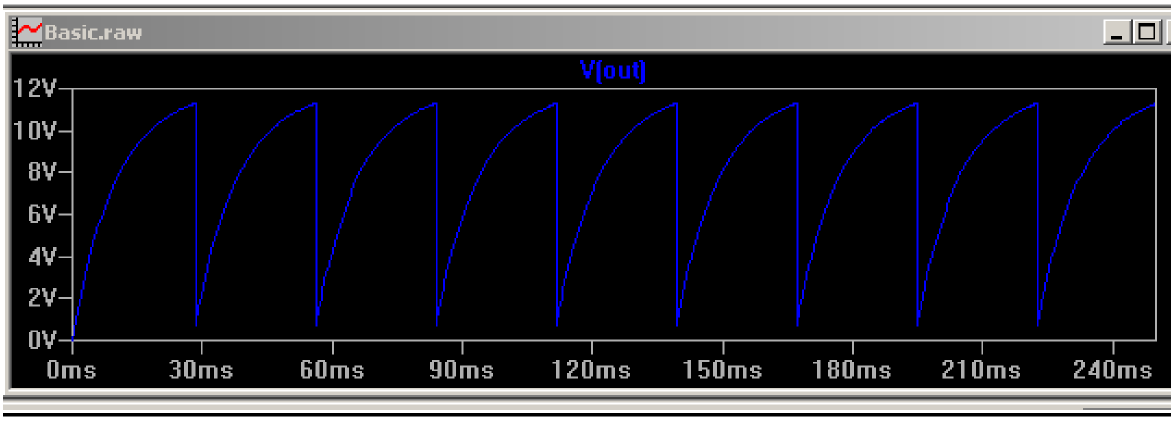
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	
Time period	
Frequency	
Bandwidth	

Result:

Thus the bandwidth of UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 5.3

WAVEFORM ANALYSIS OF UJT RELAXATION OSCILLATOR WITH DUTY CYCLE USING LTSPICE

After completion of this experiment, students will be able to:

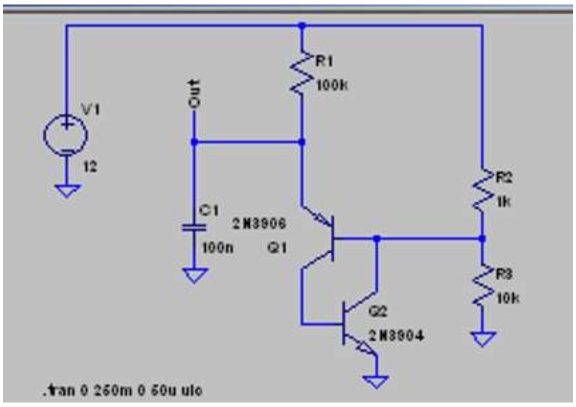
1. Understand the operating principle of a UJT Relaxation Oscillator and its frequency-determining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
6. Investigate the effect of varying resistance (R)(R)(R) and capacitance (C)(C)(C) values on the oscillation frequency.
7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
9. Apply LTSpice simulation tools for frequency analysis and performance evaluation of relaxation oscillators.

Duty Cycle

- Ratio of pulse width to total period:

$$\text{Duty Cycle} = \frac{\text{Pulse Width}}{\text{Period}} \times 100\%$$

Circuit diagram:



Procedure:

1.Connections are made as shown in the circuit diagram.

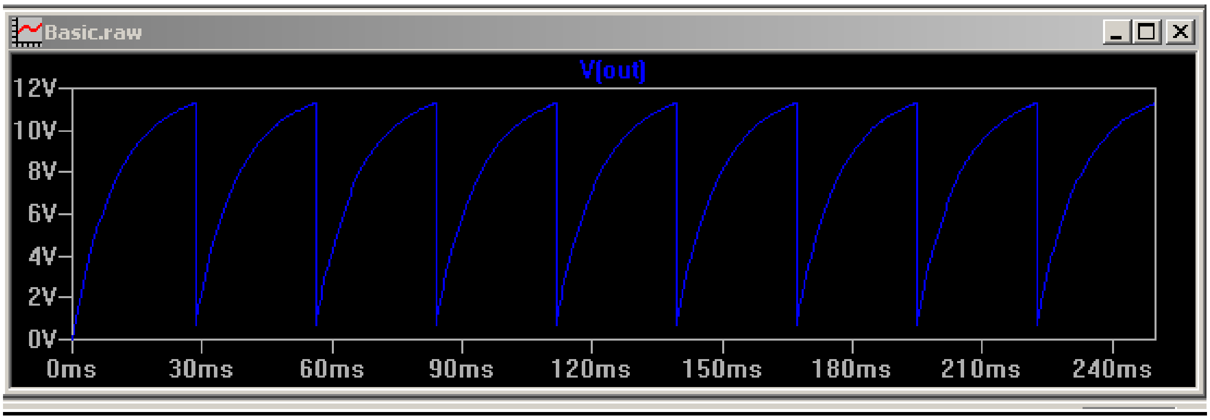
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	THEORITICAL	PRACTICAL
Pulse width		
period		
Duty Cycle		

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 6

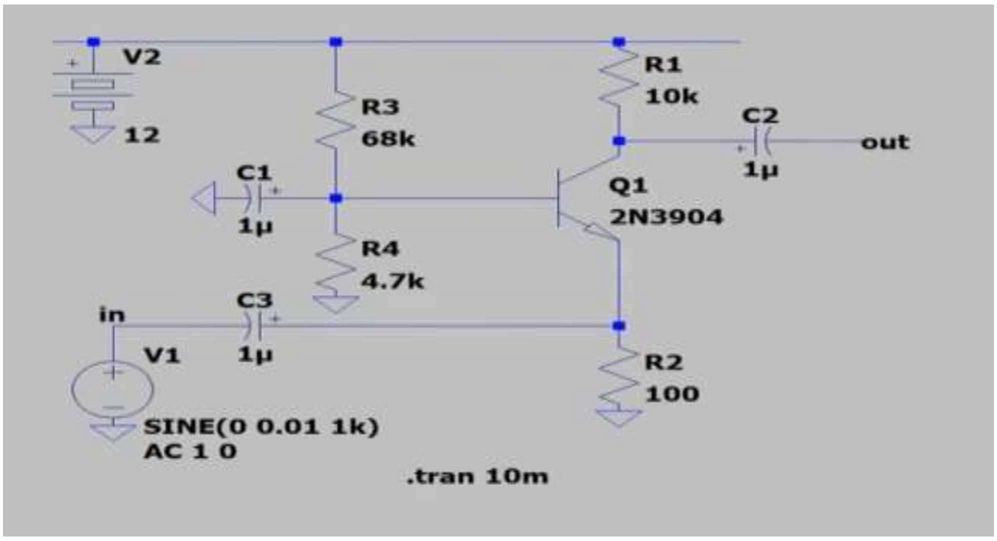
DESIGN AND SIMULATION OF COMMON BASE AMPLIFIER USING LTSPICE SIMULATOR

OBJECTIVES

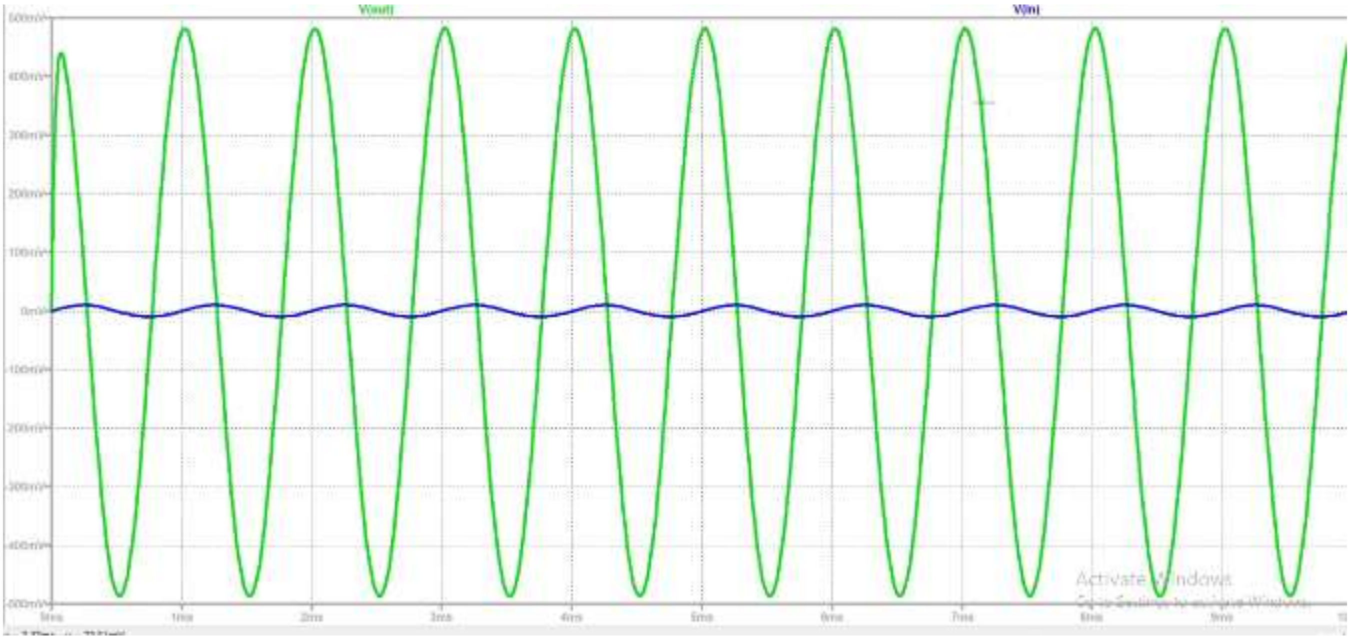
The experiment is divided into three engineering laboratory model tests:

1. Analysis of Voltage Gain
2. Input and Output Characteristic Analysis
3. Frequency Response Analysis

Circuit Diagram:



Model Graph:



EXPERIMENT – 6.1

VOLTAGE GAIN ANALYSIS OF COMMON BASE AMPLIFIER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design a Common Base amplifier using LTSpice.
2. Simulate amplifier operation.
3. Measure voltage gain.
4. Compare theoretical and simulated results.
5. Analyze CB amplifier characteristics.

Aim

To design and simulate a Common Base amplifier using LTSpice and verify its voltage gain.

THEORETICAL BACKGROUND

The **Common Base (CB) amplifier** is one of the three basic transistor amplifier configurations in which the **base terminal is common** to both the input and output circuits. In this configuration, the input signal is applied between the emitter and base terminals, while the output is taken between the collector and base terminals.

The Common Base amplifier is characterized by:

- Low input resistance
- High output resistance
- High voltage gain
- Current gain less than unity
- Excellent high-frequency performance

The CB amplifier is widely used in:

- RF amplifiers
- Impedance matching circuits
- High-frequency applications

DESIGN FORMULA:

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

Current Gain

$$A_i = \alpha$$

where

$$\alpha = \frac{I_C}{I_E}$$

EXPERIMENTAL PROCEDURE

1. Open LTSpice.
2. Create a new schematic.
3. Place NPN transistor.
4. Add biasing resistors.
5. Configure Common Base circuit.
6. Apply AC signal source.
7. Set simulation parameters.
8. Run transient analysis.
9. Measure input and output voltages.
10. Calculate gain.
11. Record observations.

TABULATION:

s.no	INPUT VOLTAGE Vi (mV)	OUTPUT VOLTAGE V0 (V)	GAIN (dB)

RESULT:

The voltage gain of the Common Base amplifier was verified successfully using LTSpice

EXPERIMENT – 6.2

MEASUREMENT OF INPUT AND OUTPUT CHARACTERISTICS OF COMMON BASE AMPLIFIER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1 Plot input characteristics.
2 Plot output characteristics.
3 Analyze transistor operation.

Aim

To obtain the input and output characteristics of a Common Base amplifier using LTSpice.

EXPERIMENTAL PROCEDURE

1 Configure DC sweep analysis.
2 Vary emitter voltage.
3 Measure emitter current.
4 Plot input characteristics.
5 Vary collector voltage.
6 Measure collector current.
7 Plot output characteristics.
8 Determine resistances.

EXPERIMENTAL OBSERVATIONS

INPUT CHARECTERISTICS

Parameter	PRACTICAL VALUE
AMPLITURE	
TIME PERIOD	

OUTPUT CHARECTERISTICS

Parameter	PRACTICAL VALUE
AMPLITURE	
TIME PERIOD	

RESULT

The input and output characteristics of the Common Base amplifier were obtained successfully.

EXPERIMENT – 6.3

FREQUENCY RESPONSE ANALYSIS COMMON BASE AMPLIFIER USING LTSPICE

The experiment is divided into three engineering laboratory model tests:

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Perform AC analysis.
2. Plot frequency response.
3. Determine bandwidth.

Aim

To obtain the frequency response characteristics of a Common Base amplifier using LTSpice.

THEORY

The Common Base amplifier exhibits excellent high-frequency response due to:

- Absence of Miller effect.
- Low input capacitance.
- Reduced feedback capacitance.

Therefore, CB amplifiers are preferred for RF applications.

DESIGN FORMULA:

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

Gain in dB

$$Gain(dB) = 20\log_{10}(A_v)$$

Bandwidth

$$BW = f_H - f_L$$

EXPERIMENTAL PROCEDURE

1 Configure AC analysis.
2 Sweep frequency from 10 Hz to 100 MHz.
3 Observe gain.
4 Plot Bode response.
5 Determine cutoff frequencies.
6 Calculate bandwidth.
7 Record observations**.**

BANDWIDTH

PARAMETER	VOUT
AMPLITUDE	
TIME PERIOD	
BANDWIDTH	

RESULT

The frequency response characteristics of the Common Base amplifier were analyzed successfully using LTSpice.

EXPERIMENT – 7

DESIGN AND SIMULATION OF FET AMPLIFIER USING LT SPICE SIMULATOR

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1.Verification of Voltage Gain

2.Determination of Lower and Upper Cutoff Frequencies

3.Frequency Response and Bandwidth Analysis

EXPERIMENT – 7.1

ANALYSIS OF VOLTAGE GAIN FOR FET AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. To design a Common Source FET amplifier
2. To simulate the amplifier circuit using LTspice.
3. Measure voltage gain experimentally.
4. Compare theoretical and practical gain values.
5. Understand transistor amplification principles.
6. Analyze the effect of transistor parameters on gain.

Aim

To design and verify the voltage gain of a FET.

Theoretical Background:

A Field Effect Transistor (FET) is a **voltage-controlled semiconductor device** having high input impedance and low power consumption.In a **Common Source Amplifier**:

- Input is applied to the gate.
- Output is taken from the drain.
- Source acts as the common terminal.
- Output is amplified and is **180° out of phase** with the input.
- The source resistor provides bias stabilization.
- The bypass capacitor increases AC gain.

The voltage gain is approximately

DESIGN FORMULAE

$$A_v = \frac{R_D}{r_s}$$

or

$$A_v = g_m(R_D||R_L)$$

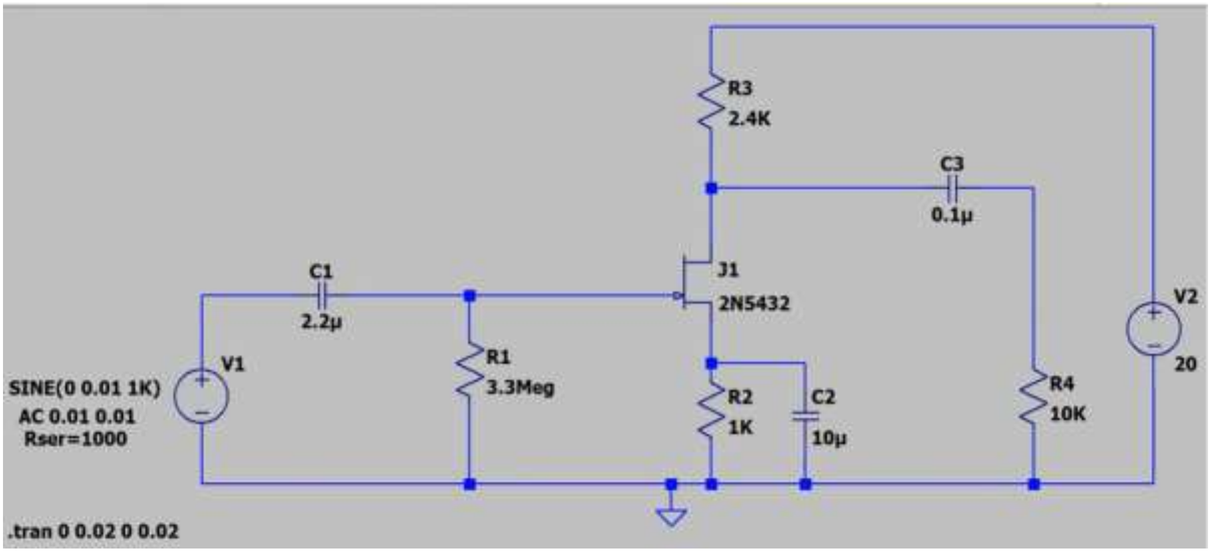
where

- g_m = Transconductance
- R_D = Drain resistance
- R_L = Load resistance

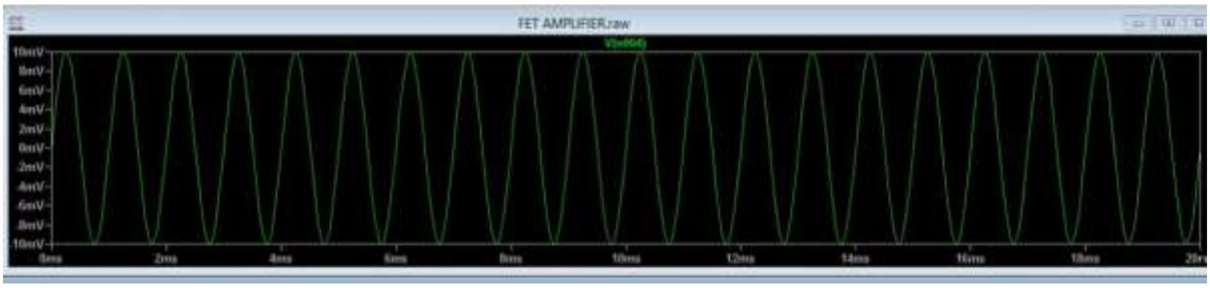
Design Procedure

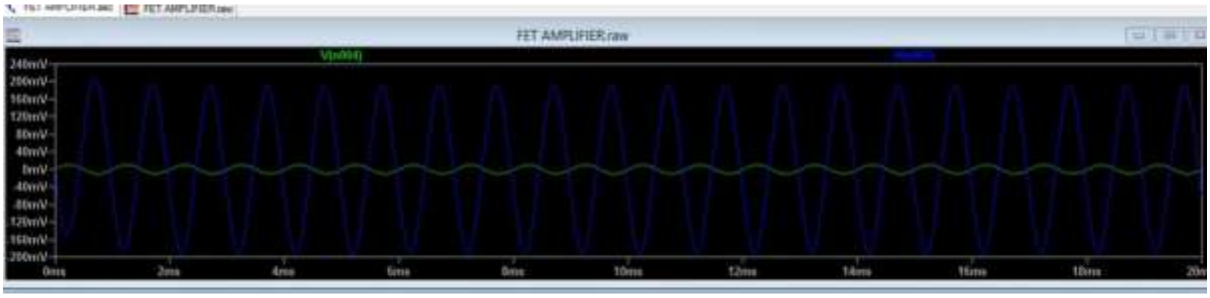
1. Select the FET device (e.g., 2N5457 or 2N7000).
2. Choose the supply voltage (typically 12 V).
3. Select RD and RS to obtain a suitable operating point.
4. Choose RG to provide proper gate bias.
5. Connect coupling and bypass capacitors.
6. Draw the circuit in LTspice.
7. Verify all component values.

Circuit diagram:



Model Graph:





Tabulation:

s.no	INPUT VOLTAGE Vi (mV)	OUTPUT VOLTAGE V0 (V)	GAIN (dB)

RESULT

The midband voltage gain of the common emitter amplifier was verified successfully.

EXPERIMENT – 7.2

DETERMINATION OF LOWER AND UPPER CUTOFF FREQUENCIES FOR FET AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Identify cutoff frequencies.
2. Calculate bandwidth.
3. Understand frequency limitations of amplifiers.
4. Interpret gain-frequency characteristics.

Aim

To determine the lower and upper cutoff frequencies of a CE amplifier.

THEORETICAL BACKGROUND

At low frequencies:

Coupling capacitor reactance increases.

Emitter bypass capacitor becomes ineffective.

Therefore gain decreases.

At high frequencies:

Internal transistor capacitances dominate.

Gain decreases due to capacitive effects.

Cutoff frequencies are defined at the points where gain becomes:

$$0.707A_{v(max)}$$

or

$$-3dB$$

from the maximum gain.

DESIGN FORMULAE

Lower Cutoff Frequency

$$f_L$$

Upper Cutoff Frequency

$$f_H$$

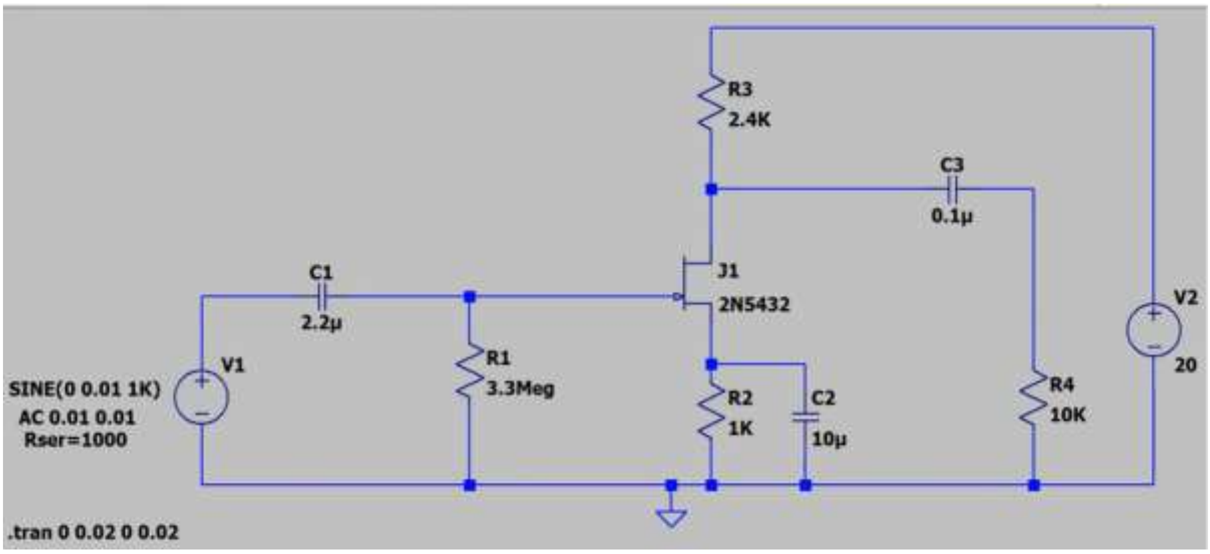
Bandwidth

$$BW = f_H - f_L$$

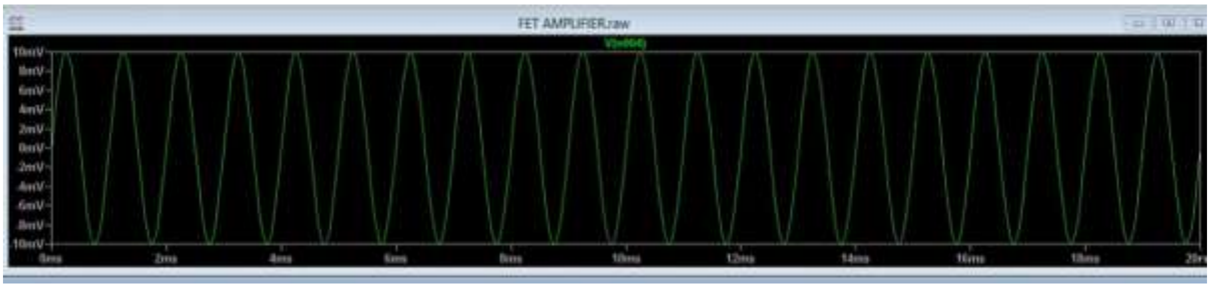
EXPERIMENTAL PROCEDURE

1. Set input voltage constant.
2. Start at 20 Hz.
3. Measure output voltage.
4. Increase frequency gradually.
5. Determine gain at each frequency.
6. Find frequencies corresponding to -3 dB points.
7. Calculate bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: V_{in}= mV

TABLE

Parameter	**Experimental Value**
LOWER CUTOFF FREQUENCY	
UPPER CUTOFF FREQUENCY	

RESULT

The lower and upper cutoff frequencies of the CE amplifier were determined successfully.

EXPERIMENT – 7.3

BANDWIDTH ANALYSIS OF COMMON EMITTER AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot frequency response characteristics.
2. Calculate bandwidth.
3. Analyze amplifier performance.
4. Construct Bode plots.

Aim

To obtain the frequency response characteristics of a CE amplifier and determine its bandwidth.

THEORY

The frequency response of an amplifier describes variation of gain with frequency.

Three frequency regions exist:

Low Frequency Region

Gain decreases due to coupling capacitors.

Mid Frequency Region

Gain remains approximately constant.

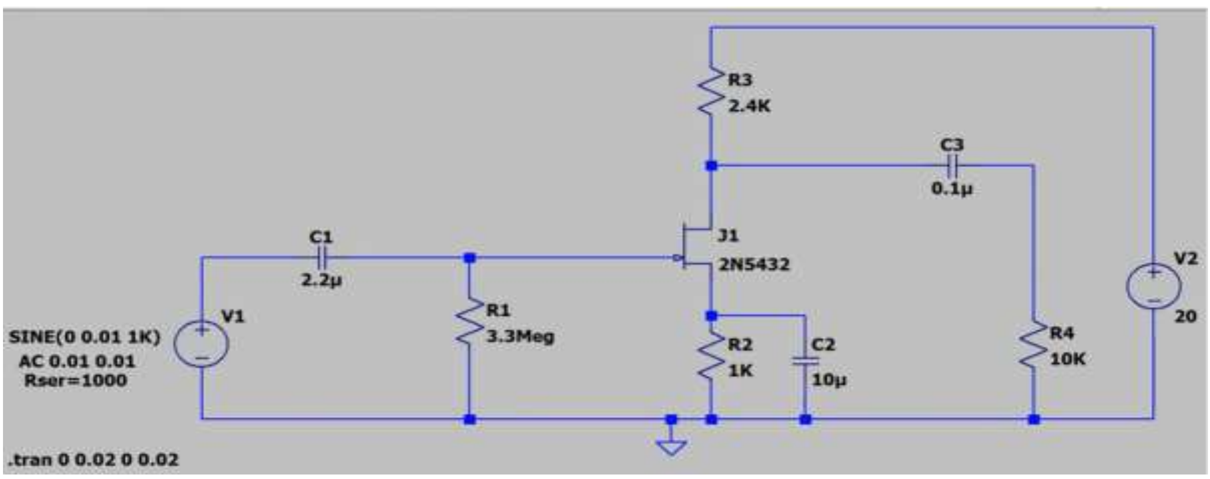
High Frequency Region

Gain decreases because of transistor junction capacitances.

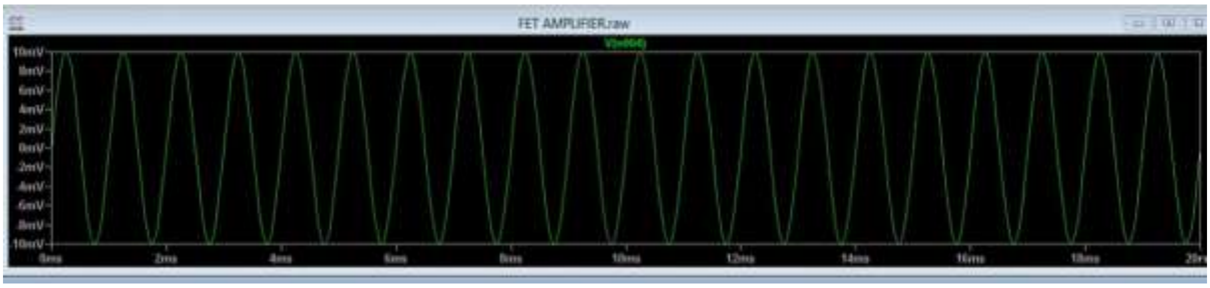
EXPERIMENTAL PROCEDURE

1. Apply constant input voltage.
2. Start at 20 Hz.
3. Record output voltage.
4. Increase frequency gradually.
5. Continue measurements up to 1 MHz.
6. Calculate gain.
7. Convert gain into dB.
8. Plot Bode magnitude response.
9. Determine bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: $\mathbf{V_{in}}$**= 20 mV**

TABLE

Parameter	**Experimental Value**
LOWER CUTOFF FREQUENCY	
UPPER CUTOFF FREQUENCY	
BANDWIDTH	

RESULT

The frequency response characteristics of the common emitter amplifier were obtained successfully and the bandwidth was determined.

EXPERIMENT – 8

DESIGN AND ANALYSIS OF ASTABLE MULTIVIBRATOR USING BJT

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1. Switching Analysis Astable Multivibrator
2. Measurement of Time Period and Frequency
3. Duty Cycle Analysis

EXPERIMENT – 8

.1

SWITCHING ANALYSIS OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an astable multivibrator using BJT.
2. Observe continuous switching operation.
3. Analyze transistor switching behavior.
4. Generate square wave signals.
5. Compare theoretical and practical waveforms.

Aim

To design and verify the operation of an astable multivibrator using BJT.

THEORETICAL BACKGROUND

An astable multivibrator is a free-running oscillator having no stable state. The circuit continuously switches between two quasi-stable states due to regenerative feedback.

Characteristics:

- No external triggering required.
- Generates square wave output.
- Used as clock pulse generator.
- Produces continuous oscillations.

During operation:

- One transistor conducts while the other remains OFF.
- Capacitors charge and discharge alternately.
- Switching occurs automatically.

DESIGN FORMULA:

Time Period

$$T = 1.38RC$$

Frequency

$$f = \frac{1}{T}$$

ON Time

$$T_1 = 0.69R_1C_1$$

OFF Time

$$T_2 = 0.69R_2C_2$$

2. Frequency of Oscillation

$$f = \frac{1}{T}$$

Substituting T:

$$f = \frac{1}{1.386RC}$$

Approximation:

$$f \approx \frac{0.72}{RC}$$

3. Capacitor Calculation

$$C = \frac{1}{1.386 f R}$$

4. Resistor Calculation

$$R = \frac{1}{1.386 f C}$$

5. ON Time of Transistor Q1

$$t_{ON1} = 0.693 R_2 C_1$$

6. ON Time of Transistor Q2

$$t_{ON2} = 0.693 R_3 C_2$$

7. Total Time Period

$$T = t_{ON1} + t_{ON2}$$

$$T = 0.693(R_2 C_1 + R_3 C_2)$$

8. Frequency (General Case)

$$f = \frac{1}{0.693(R_2C_1 + R_3C_2)}$$

9. Duty Cycle

$$D = \frac{t_{ON}}{T} \times 100$$

For symmetrical operation:

$$D = 50\%$$

when

$$R_2 = R_3$$

and

$$C_1 = C_2$$

10. Collector Current

$$I_C = \beta I_B$$

11. Base Current

$$I_B = \frac{I_C}{\beta}$$

12. Base Resistor

$$R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

13. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE(sat)}}{I_C}$$

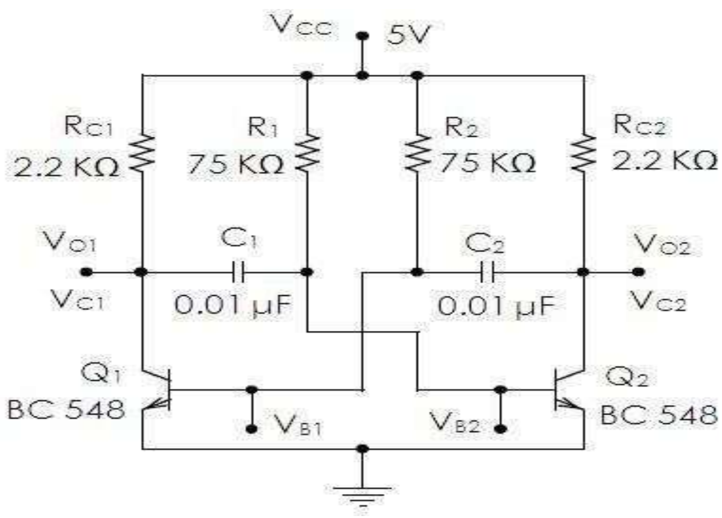
14. Output Peak-to-Peak Voltage

$$V_{pp} \approx V_{CC}$$

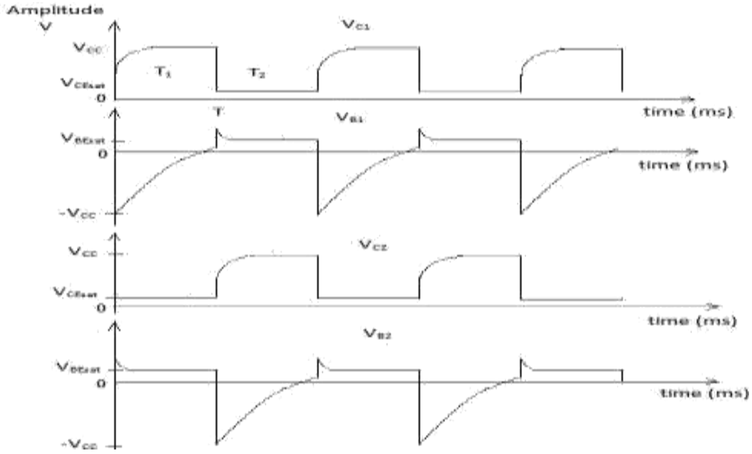
EXPERIMENTAL PROCEDURE

1. Verify all components.
2. Assemble the astable multivibrator circuit.
3. Apply DC supply voltage.
4. Observe collector and base outputs of both transistors.
5. Verify alternate switching.
6. Observe square wave output.
7. Measure time period.
8. Record waveform characteristics.

CIRCUIT DIAGRAM:



MODEL GRAPH



Tabulation:

S.no	Amplitude (Volts)	TON (ms)	TOFF (ms)
Theoretical value			
Practical Value			

Result:

Astable Multivibrator was designed and frequency of oscillation was calculated.

Frequency of oscillation f= ________(Hz)

EXPERIMENT – 8.2

MEASUREMENT OF TIME PERIOD AND FREQUENCY OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Measure time period experimentally.
2. Determine oscillation frequency.
3. Verify theoretical calculations.
4. Analyze capacitor charging and discharging effects.

Aim

To measure the time period and frequency of an astable multivibrator.

THEORETICAL BACKGROUND

The oscillation frequency depends on resistor and capacitor values. Capacitors alternately charge and discharge through base resistors causing periodic switching.

For symmetrical operation:

$$T = 1.38RC$$

Frequency:

$$f = \frac{1}{T}$$

Increasing R or C increases time period and decreases frequency.

Time Period $T=T_1+T_2$ where T_1= ON time T_2= OFF time

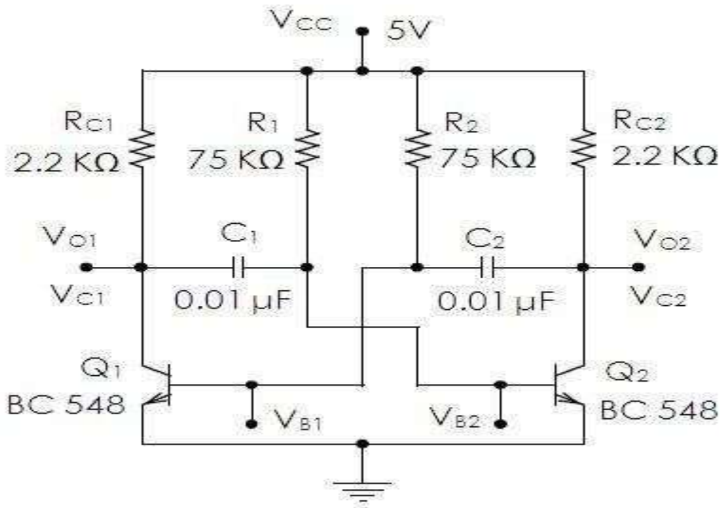
Frequency f=1/ T in Hz.

EXPERIMENTAL PROCEDURE

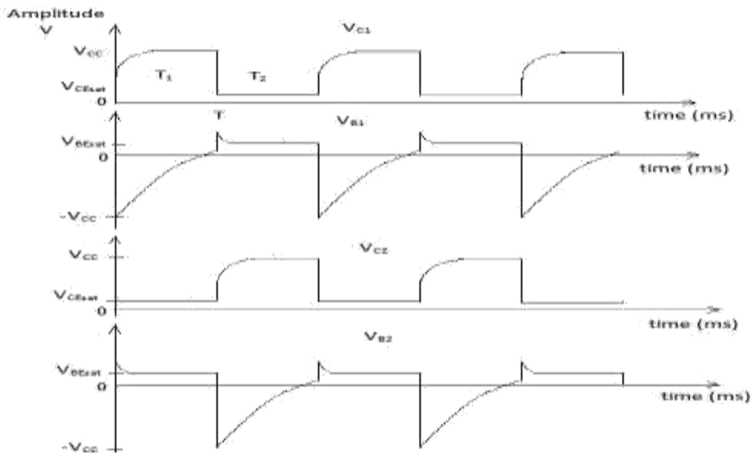
1. Assemble the astable multivibrator.
2. Measure waveform using CRO.
3. Determine one complete cycle.
4. Measure time period.
5. Calculate frequency.

6. Compare theoretical and practical values.
7. Record observations.

Circuit diagram:



Model Graph:



Procedure:

1. Test the components.
2. Assemble the circuit on a breadboard.
3. Switch ON the power supply.
4. Connect the outputs of the circuit to an oscilloscope.
5. Observe the **collector** and **base** waveforms of the two transistors.
6. Measure the **frequency** and **amplitude** of the outputs.
7. Plot the waveforms.

Tabulation:

S.no	Amplitude (Volts)	TON (ms)	TOFF (ms)	Frequency (Hz)
Theoretical value				
Practical Value				

RESULT

The time period and frequency of the astable multivibrator were measured successfully.

EXPERIMENT – 8.3

DUTY CYCLE ANALYSIS OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Measure ON time and OFF time.
2. Calculate duty cycle.
3. Analyze waveform symmetry.
4. Evaluate pulse generation characteristics.

Aim

To determine the duty cycle of the astable multivibrator.

THEORY

Duty cycle is defined as the ratio of ON time to total time period.

Duty Cycle

$$D = \frac{T_{ON}}{T} \times 100$$

where

$$T = T_{ON} + T_{OFF}$$

For symmetrical astable multivibrators:

$$D \approx 50\%$$

DESIGN FORMULA:

ON Time

$$T_{ON} = 0.69RC$$

OFF Time

$$T_{OFF} = 0.69RC$$

Duty Cycle

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$$

EXPERIMENTAL PROCEDURE

1. Observe output waveform on CRO.
2. Measure ON duration.
3. Measure OFF duration.
4. Calculate total period.
5. Determine duty cycle.
6. Compare with theoretical value.
7. Record observations.

Duty Cycle Measurement

Test Case	**TON (ms)**	**TOFF (ms)**	**Time Period T (ms)**	**Duty Cycle (%)**

RESULT

The duty cycle of the astable multivibrator was determined successfully and verified with theoretical calculations.

EXPERIMENT – 9

DESIGN AND ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

OBJECTIVES

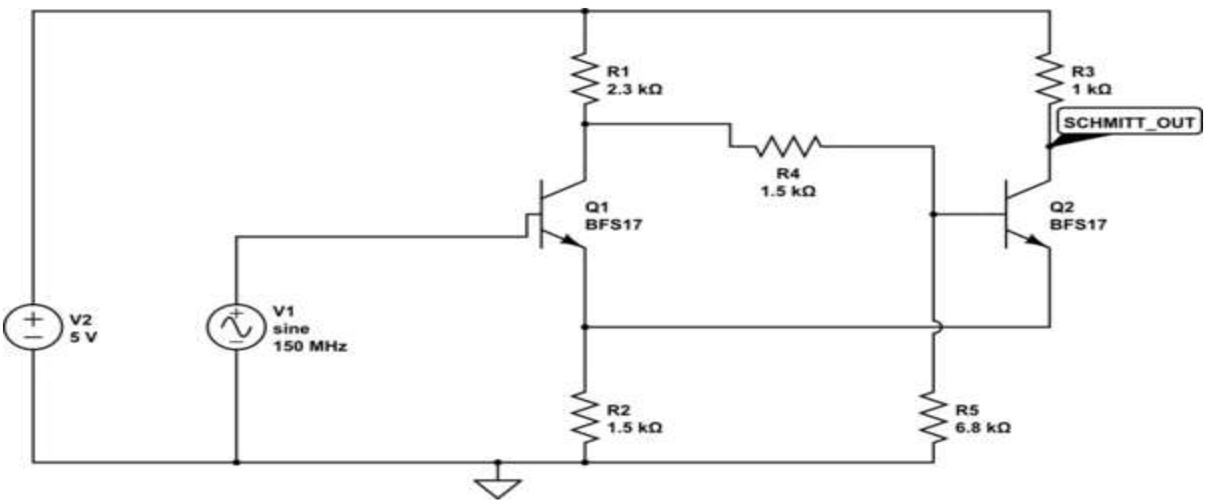
The experiment is divided into three engineering laboratory model tests:

1.Analyse function of Schmitt trigger and determine bandwidth

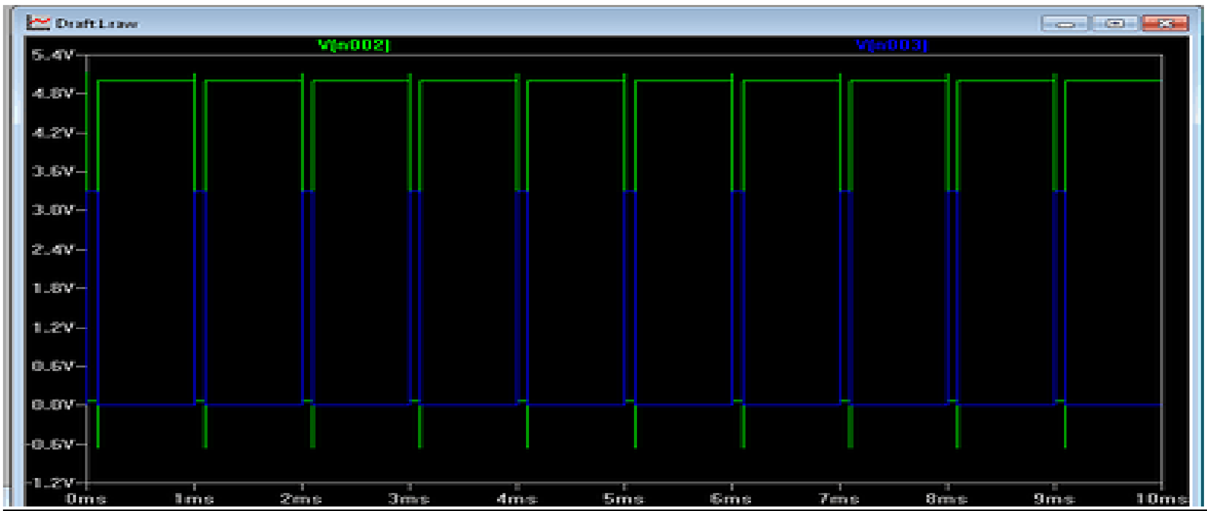
2.Determination Of UTP and LTP Frequency Response Analysis

3.Analysis of Hysteresis Characteristics

CIRCUIT DIAGRAM:



Model Graph:



EXPERIMENT – 9.1

BANDWIDTH ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

- Understand regenerative feedback.
- Analyze bistable switching action.
- Measure trigger levels.
- Determine bandwidth

Aim

To simulate and verify the operation of a Schmitt Trigger and determine bandwidth using LTSpice.

THEORY

A Schmitt Trigger is a regenerative comparator circuit exhibiting hysteresis.

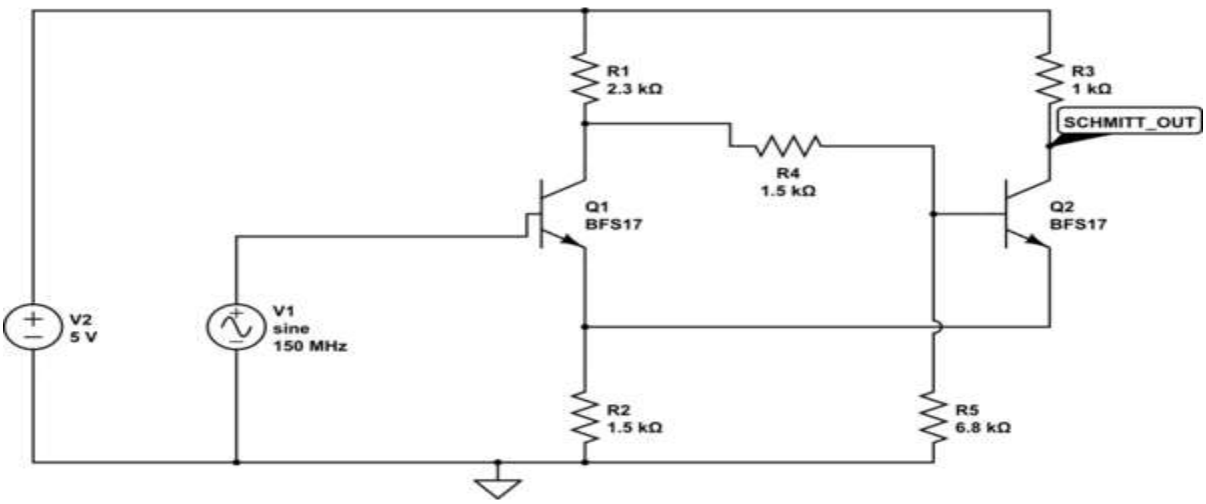
It has:

- Two stable states.
- Two switching thresholds.
- Excellent noise immunity.

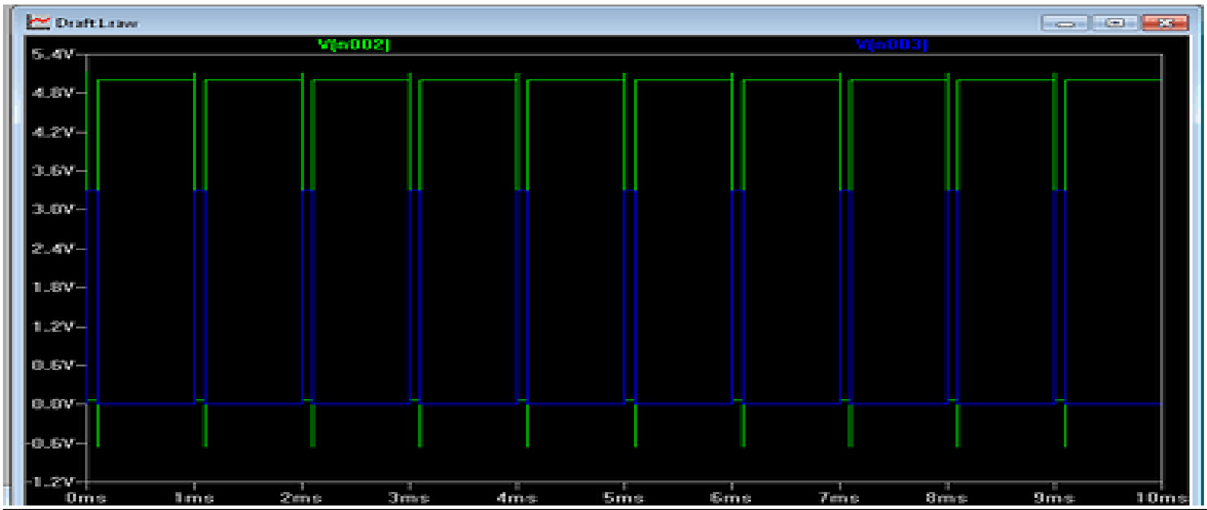
Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

CIRCUIT DIAGRAM:



Model Graph:



BANDWIDTH

PARAMETER	VOUT
AMPLITUDE	
TIME PERIOD	
BANDWIDTH	

Result:

Schmitt Trigger using LTSpice was completed and output was obtained.

EXPERIMENT – 9.2

DETERMINATION OF UTP AND LTP FOR SCHMITT TRIGGER USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

Aim

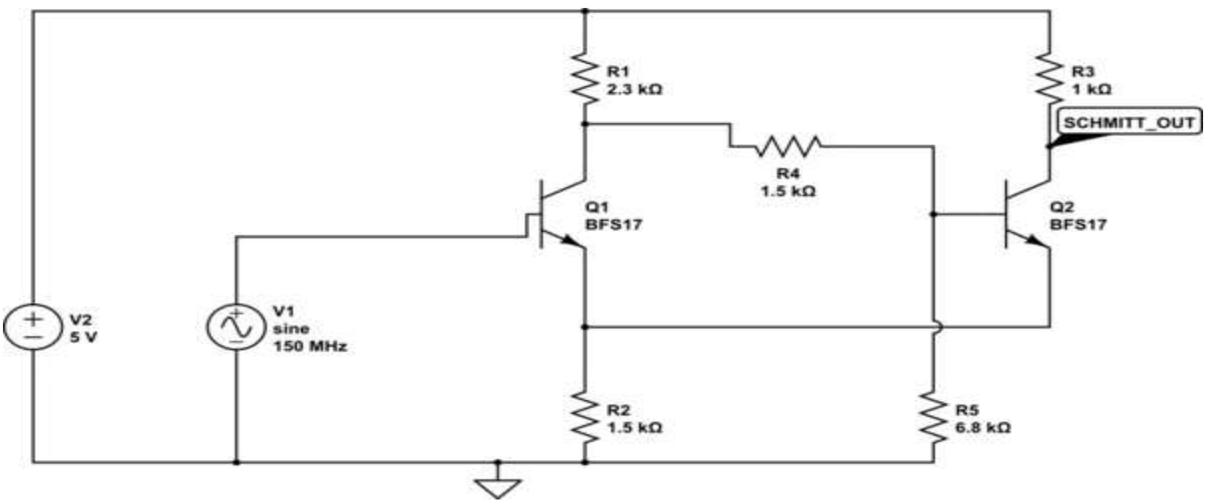
To determine Upper Trigger Point and Lower Trigger Point for Schmit trigger circuit

DESIGN FORMULA:

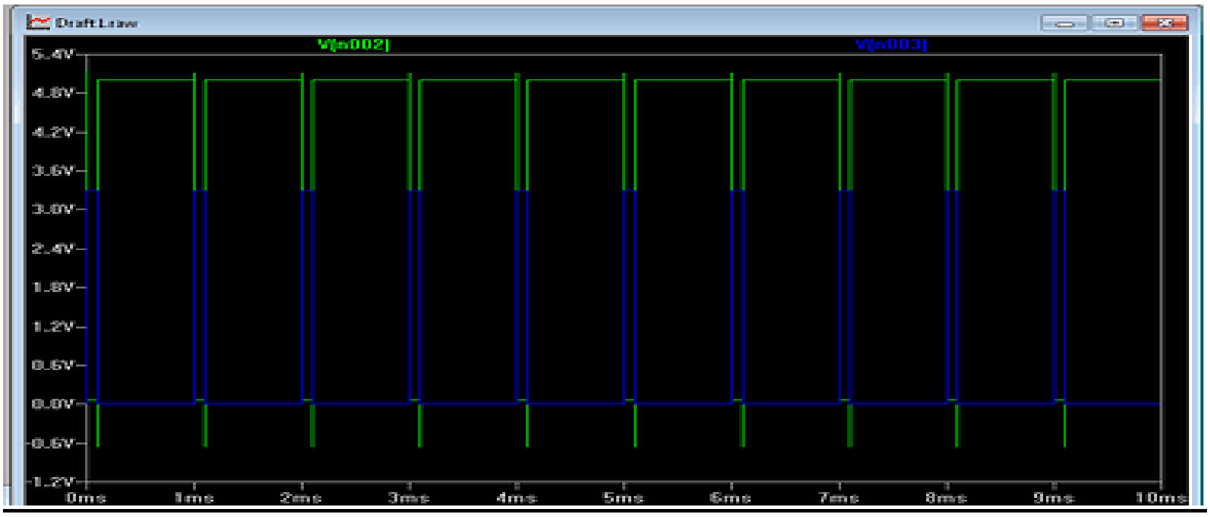
Hysteresis Width

$$V_H = UTP - LTP$$

CIRCUIT DIAGRAM:



Model Graph:



TABULATION

PARAMETER	VOUT
UTP	
LTP	
HYSTERESIS WIDTH	

RESULT

The trigger levels were measured successfully.

EXPERIMENT – 9.3

ANALYSIS OF HYSTERESIS CHARACTERISTICS FOR SCHMITT TRIGGER USING LTSPICE

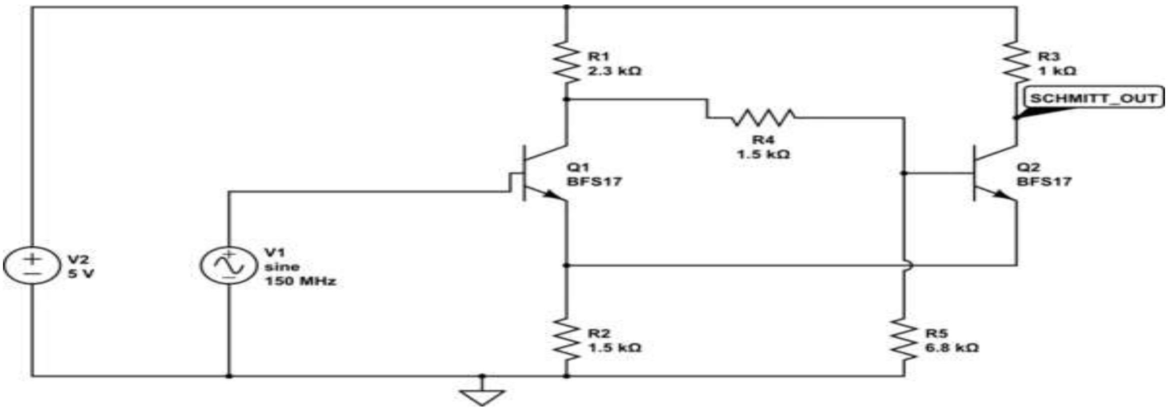
After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Observe and interpret the switching characteristics of the Schmitt Trigger.
7. Plot and analyze the hysteresis transfer characteristic curve.
8. Evaluate the noise immunity provided by hysteresis in comparator circuits.
9. Compare the operation of a Schmitt Trigger with that of a conventional comparator.
10. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

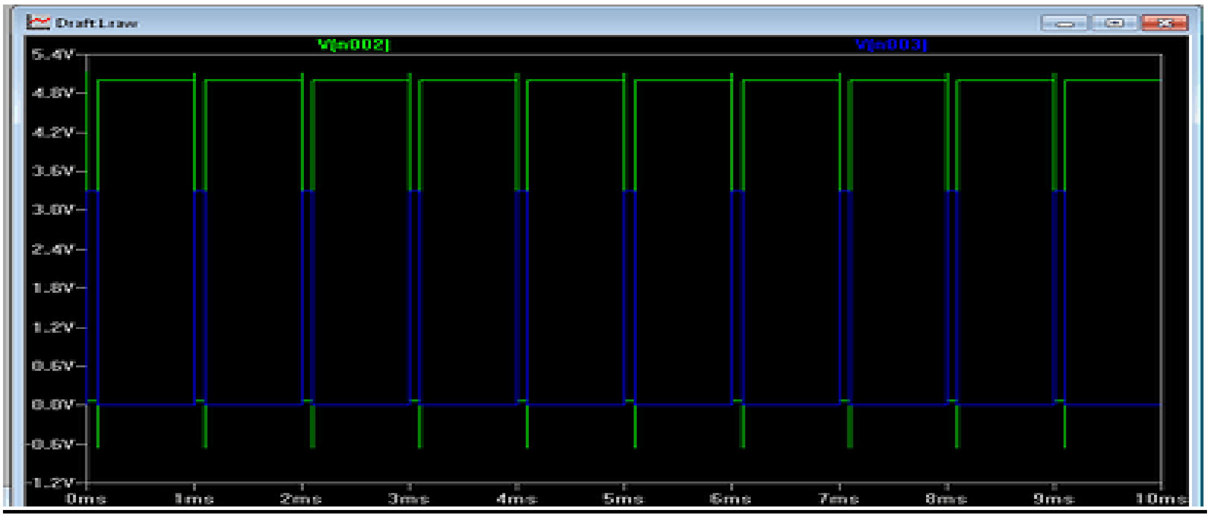
Aim

To plot hysteresis characteristics of a Schmitt Trigger.

CIRCUIT DIAGRAM:



<u>**Model Graph:**</u>



<u>**TABULATION**</u>

PARAMETER	VOUT
VIN	
VOUT	

RESULT:

Thus the plot hysteresis characteristics of a Schmitt Trigger was done.

EXPERIMENT – 10

Design of Hartley Oscillator Using PCB Software

OBJECTIVES

1. Design and Create PCB layout for Hartley Oscillator circuit.
2. Route PCB tracks with Manufacturing board
3. Generate fabrication files.

Expected Learning Outcomes

After completion of this experiment, students will be able to:

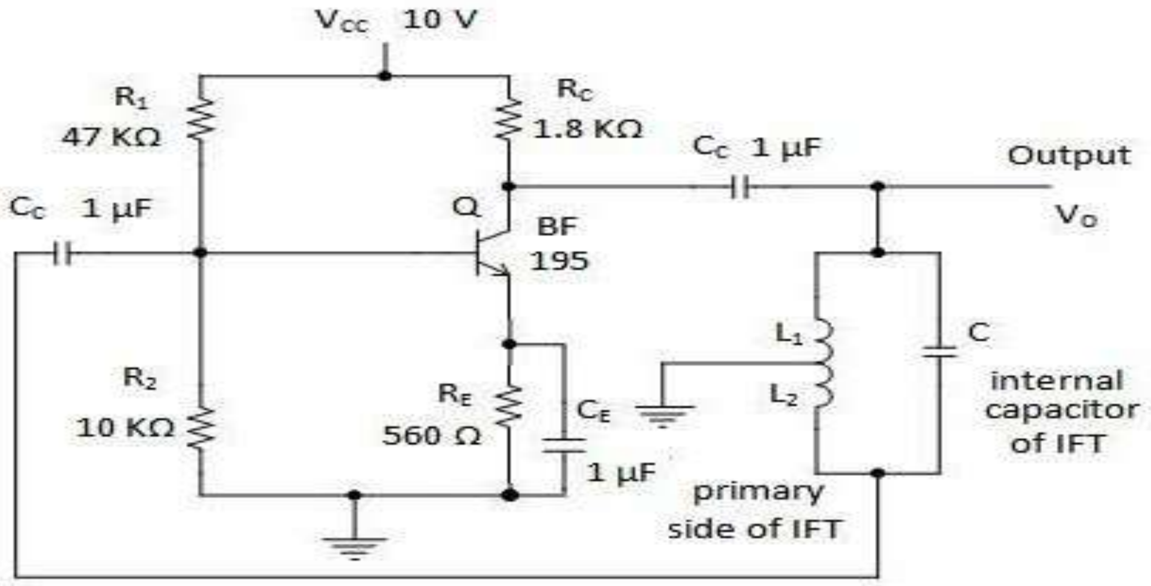
1. Understand the operating principle of a Hartley Oscillator and its applications.
2. Design the schematic of a Hartley Oscillator using EAGLE CAD software.
3. Calculate the oscillation frequency using inductance and capacitance values.
4. Create a PCB layout by placing components and routing tracks effectively.
5. Perform Design Rule Check (DRC) to verify PCB manufacturability.
6. Generate Gerber and drill files required for PCB fabrication.
7. Analyze PCB design considerations such as component placement, grounding, and signal routing.
8. Apply modern PCB design tools for analog circuit implementation.

Theory

A Hartley oscillator is an LC oscillator that uses a tapped inductor for feedback. The total inductance of the tank circuit determines the frequency of oscillation. Positive feedback is achieved through inductive coupling. The condition for sustained oscillations is given by the Barkhausen criterion.

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

Circuit Diagram



EXPERIMENT 10.1

DESIGN A HARTLEY OSCILLATOR CIRCUIT USING PCB SOFTWARE AND IMPLEMENT ITS PCB LAYOUT

Aim

To design a Hartley oscillator circuit and implement its PCB layout using EAGLE software

Apparatus / Software Required

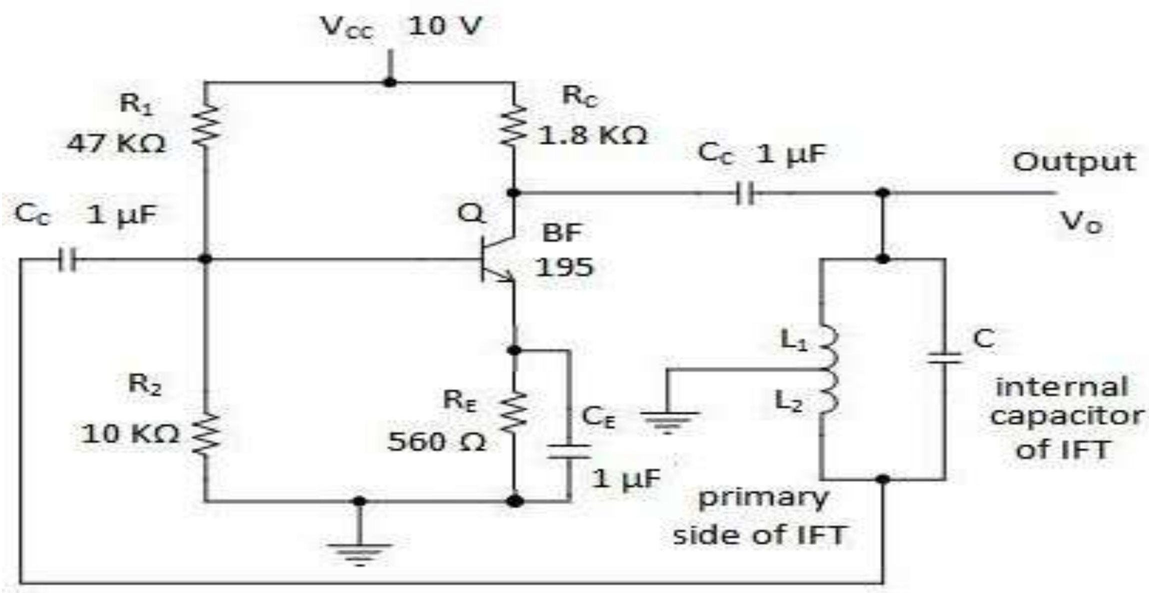
- PCB CAD Software
- PC

Theory

A Hartley oscillator is an LC oscillator that uses a tapped inductor for feedback. The total inductance of the tank circuit determines the frequency of oscillation. Positive feedback is achieved through inductive coupling. The condition for sustained oscillations is given by the Barkhausen criterion.

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

Circuit diagram:

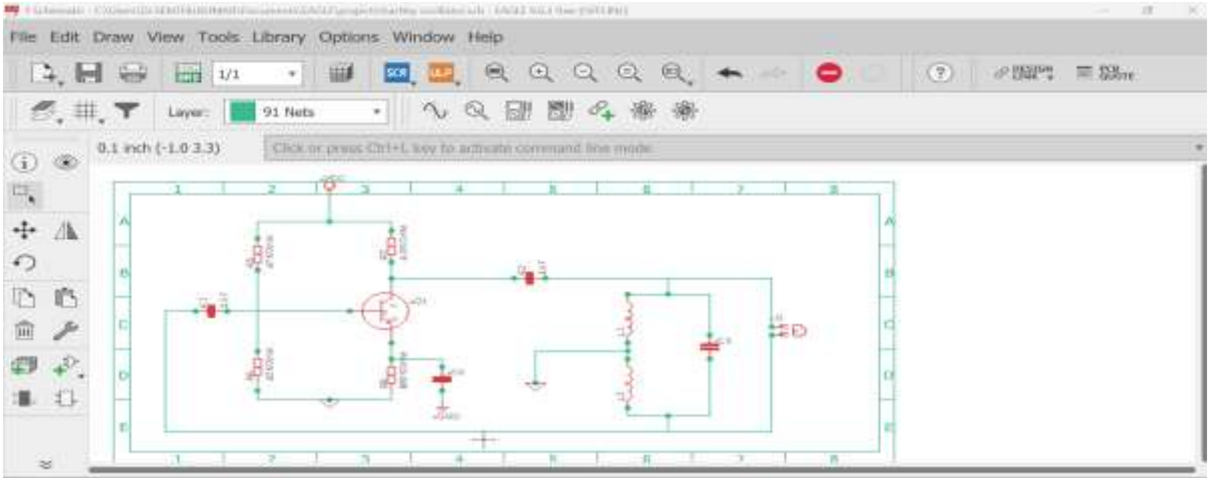


Procedure

Schematic Design

- A new project is created in EAGLE.
- Components such as BF195, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

PCB SCHEMATIC DIAGRAM



Result

The Hartley oscillator Schematic diagram was successfully designed using PCB software.

EXPERIMENT – 10.2

ROUTE PCB TRACKS WITH MANUFACTURING BOARD FOR HARTLEY OSCILLATOR CIRCUIT USING EAGLE SOFTWARE

Aim

To **route PCB tracks with manufacturing board for** monostable multivibrator schematic using Eagle Software

Apparatus / Software Required

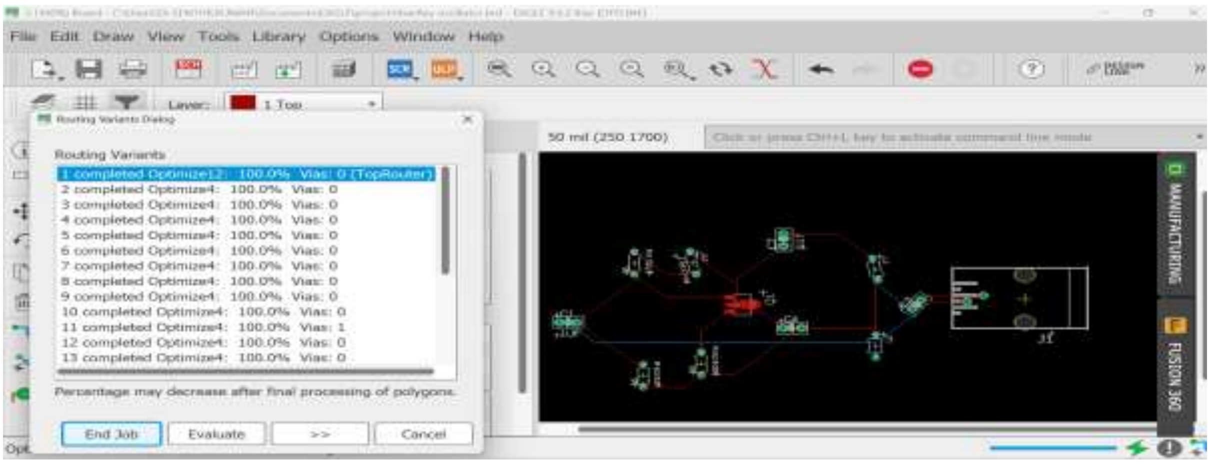
- PCB CAD Software
- PC

Procedure

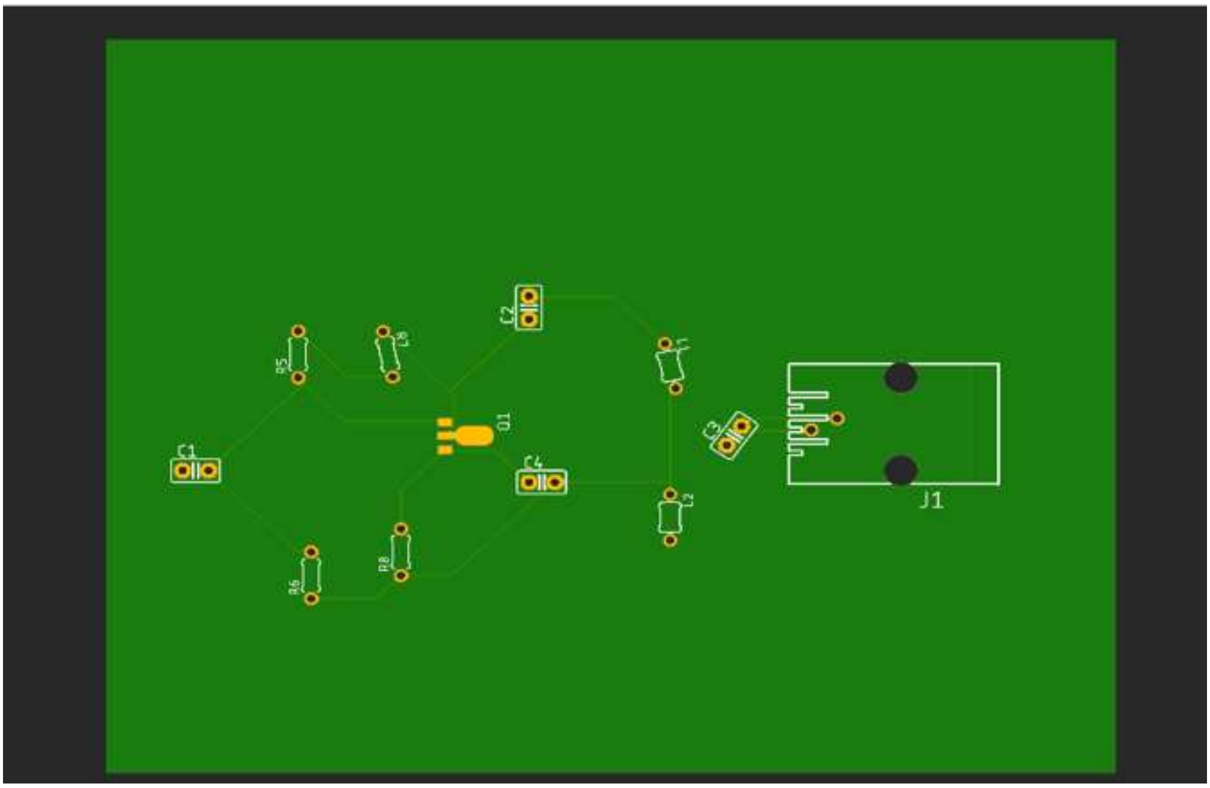
PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT



RESULT

Thus, **Routing and PCB tracks with manufacturing board for** monostable multivibrator designed successfully.

EXPERIMENT – 10.3

DESIGN OF HARTLEY OSCILLATOR TO GENERATE GERBER FILES USING EAGLE SOFTWARE

Aim

To design a Hartley oscillator circuit and implement its PCB layout using EAGLE software, and to generate Gerber files for fabrication.

Apparatus / Software Required

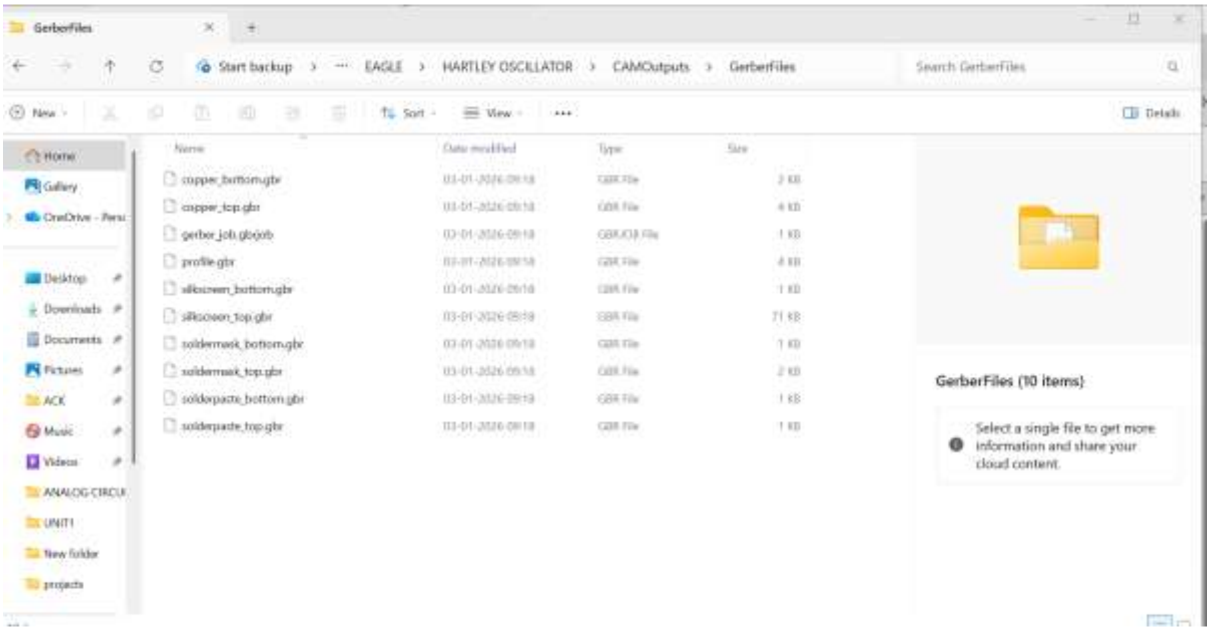
- PCB CAD Software
- Transistor
- Resistors, Capacitors, Inductors
- DC Power Supply (10 V)
- Gerber Viewer

Procedure:

Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

GERBER FILE DETAILS



Result

The Hartley oscillator circuit was successfully designed using PCB software. The PCB layout was completed and Gerber files were generated for fabrication.

EXPERIMENT – 11

Design of Monostable Multivibrator Using PCB Software

OBJECTIVES

1. Design and Create PCB layout Monostable Multivibrator circuit.
2. Route PCB tracks with Manufacturing board
3. Generate fabrication files.

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Understand the operation and applications of a Monostable Multivibrator.
2. Design the schematic of a Monostable Multivibrator using EAGLE CAD software.
3. Calculate pulse width using circuit parameters and timing equations.
4. Develop a PCB layout for the monostable circuit following standard design practices.
5. Perform component placement and PCB routing for reliable circuit operation.
6. Verify the PCB design using Design Rule Check (DRC).
7. Generate Gerber and drill files for PCB manufacturing.
8. Apply PCB design and fabrication concepts for practical implementation of pulse generation circuits.

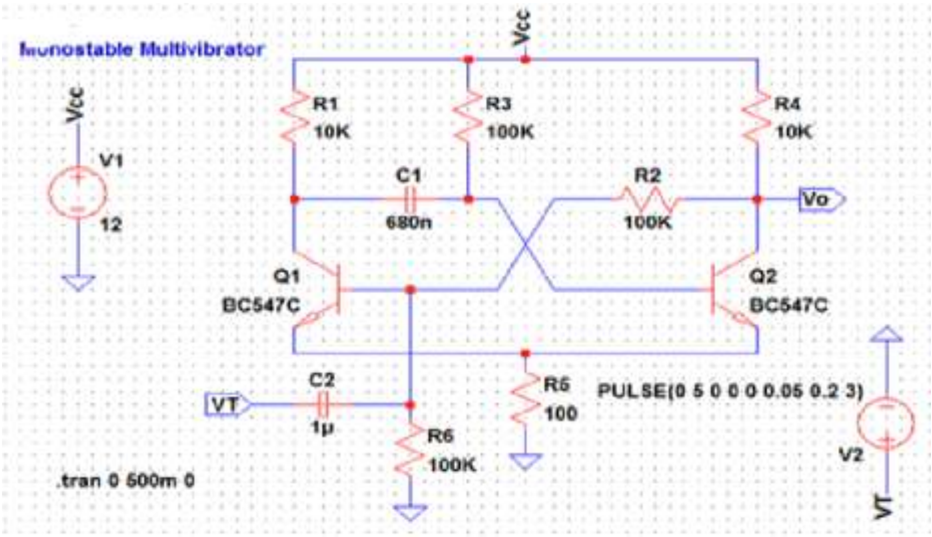
Theory

A monostable multivibrator has **one stable state and one quasi-stable state**.
When a trigger pulse is applied, the circuit switches to the quasi-stable state for a fixed duration and then returns to the stable state automatically.

PulseWidth Equation

T=0.69×R×CT = 0.69 \times R \times CT=0.69×R×C

Circuit Diagram



Experiment 1.1

DESIGN A MONOSTABLE MULTIVIBRATOR CIRCUIT AND IMPLEMENT ITS PCB LAYOUT USING EAGLE SOFTWARE

Aim

To design a **Monostable Multivibrator** circuit and implement its PCB layout using EAGLE software

Apparatus / Software Required

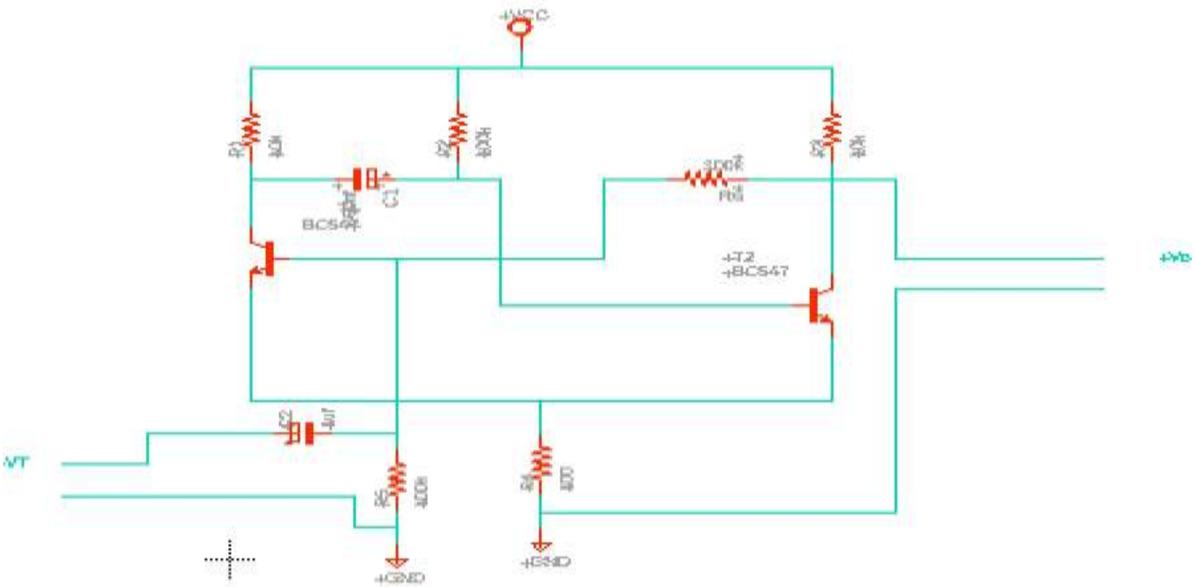
- PCB CAD Software
- PC

Procedure

Schematic Design

- A new project is created in EAGLE.
- Components such as BC547, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

PCB SCHEMATIC DIAGRAM



Result

The monostable multivibrator Schematic diagram was successfully designed using EAGLE software

EXPERIMENT – 11.2

ROUTE PCB TRACKS WITH MANUFACTURING BOARD FOR MONOSTABLE MULTIVIBRATOR USING EAGLE SOFTWARE

Aim

To **route PCB tracks with manufacturing board for** monostable multivibrator schematic using PCB Software

Apparatus / Software Required

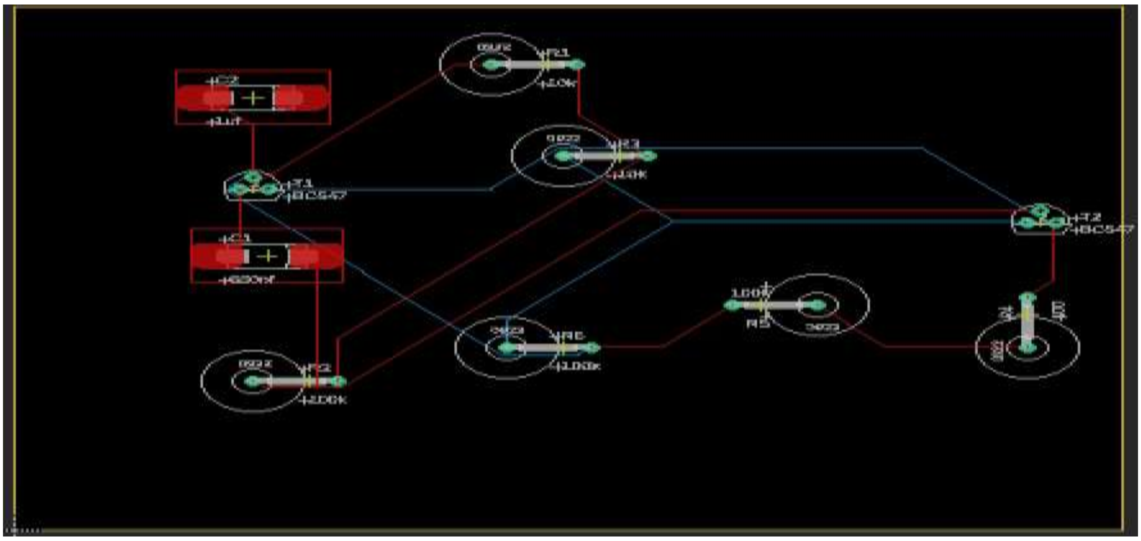
- PCB CAD Software
- PC

Procedure

PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT

RESULT

Thus, **Routing and PCB tracks with manufacturing board for** monostable multivibrator designed successfully.

EXPERIMENT – 11.3

DESIGN OF MONOSTABLE MULTIVIBRATOR TO GENERATE GERBER FILES USING EAGLE SOFTWARE

Aim

To design a monostable multivibrator circuit and implement its PCB layout using EAGLE software, and to generate Gerber files for fabrication.

Apparatus / Software Required

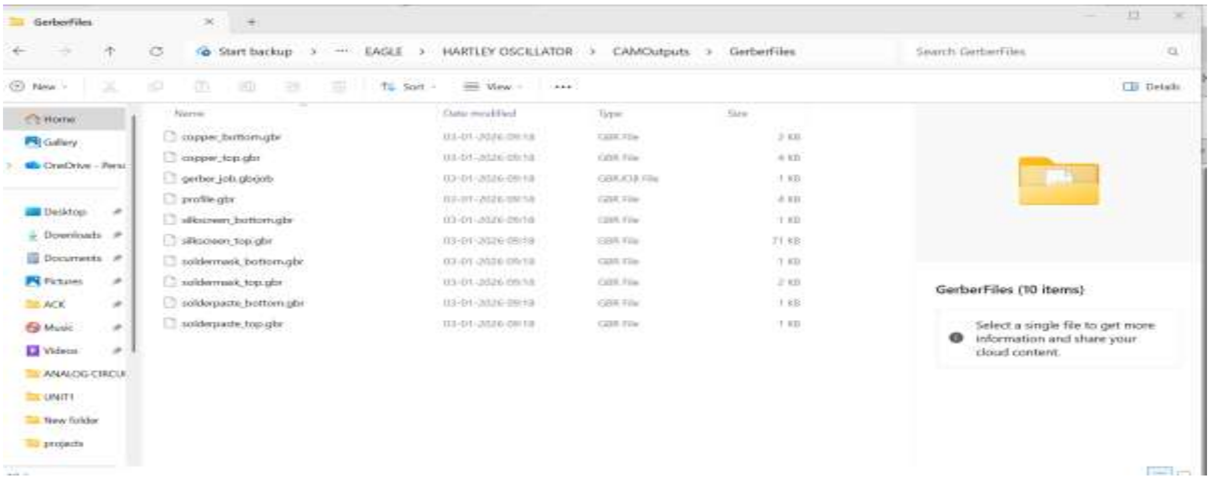
- PCB CAD Software
- PC

Procedure:

Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

GERBER FILE DETAILS



Result

The Hartley oscillator circuit was successfully designed using PCB software. The PCB layout was completed and Gerber files were generated for fabrication.